

Bachelor Thesis

Exploring and interpreting the near-optimal space of the
European energy system

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Abstract

Energy-system optimisation models compute least-cost designs of future energy systems, but much of what shapes real decisions, from public acceptance to political feasibility, lies outside these models, and realised transitions deviate from the computed optimum. The near-optimal space, the designs within a small cost margin of the optimum, holds the alternatives that such unmodelled factors may favour; Modelling to Generate Alternatives explores this space, yet its established methods cannot state how much of it they leave unexplored. ORACLE closes this gap: it encloses the near-optimal space between an inner and an outer approximation and certifies the maximum error of reading the space off either one, so far demonstrated in six dimensions. This thesis asks whether ORACLE converges on a large sector-coupled model of the European energy system, and what structure the explored near-optimal space has. We apply the method to a 28-node greenfield model of the 2050 system, exploring nine debated technology and import decisions and total cost as a tenth coordinate within a cost slack of 10%, and characterise the space through certified necessary conditions, uniform interior samples and pairwise correlations. The exploration itself was robust, steadily refining the two approximations, but certifying the achieved coverage became the bottleneck: the final certificate places the coverage error between 0.085 and 0.202 in normalised units, straddling the target tolerance of 0.1, so convergence was neither certified nor refuted. Within the explored space, no single technology or import is required, yet every near-optimal design builds at least 1480 GW of variable renewable capacity and 15.6 ktCO₂eq/h of CO₂-capture capacity, and most of the near-optimal volume lies well inside these certified ranges, with the technologies at most moderately coupled. The European 2050 system is thus constrained in its aggregates but open in its technology mix: the required renewable and capture totals hold across every near-optimal design, whereas which technologies deliver them is left open, to be settled on grounds the model does not represent. Even without reaching the tolerance, the certification still bounds how much of the space the exploration may have missed, so its coverage is stated rather than assumed.

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List of Abbreviations

Acronyms

CCS carbon capture and storage

DAC direct air capture

ESS effective sample size

HSJ Hop-Skip-Jump

KKT Karush–Kuhn–Tucker

KSG Kraskov–Stögbauer–Grassberger

LP linear program

MAA Modelling All Alternatives

MCMC Markov-chain Monte Carlo

MGA Modelling to Generate Alternatives

MILP mixed-integer linear program

OLS ordinary least squares

ORACLE Optimization for the Rigorous Analysis of the Complete Landscape of alternatives in Energy systems

PCA principal-component analysis

PV photovoltaics

QP quadratic program

SOS1 special ordered set of type 1

TSA time-series aggregation

VMM variable min/max

VRE variable renewable energy

List of Mathematical Symbols

Sets and spaces

$\mathcal{X}$	Feasible space of the model (all investment and operation variables)
$\mathcal{X}_\varepsilon$	Near-optimal space (feasible designs with cost within $(1 + \varepsilon) C^*$)
$\mathcal{Z}_\varepsilon$	Near-optimal space in exploratory coordinates
$\mathcal{I}, \mathcal{O}$	Inner and outer approximation of $\mathcal{Z}_\varepsilon$
$\mathcal{M}_i$	Member set (technologies or carriers) of axis i
$\mathcal{N}, \mathcal{T}$	Node set; set of representative time steps

Vectors and matrices

$\mathbf{x}$	Model decision vector (n_x variables)
$\mathbf{c}$	Cost coefficient vector
$\mathbf{w}$	MGA weight vector
$\mathbf{z}, \bar{\mathbf{z}}$	Exploratory variables; normalised exploratory variables
$\mathbf{S}$	Aggregation map, $\mathbf{z} = \mathbf{S}\mathbf{x}$
$\mathbf{Z}$	Matrix of the m certified designs (rows)
$\boldsymbol{\lambda}$	Convex-combination weights
$\mathbf{A}, \mathbf{b}$	Half-space description of $\mathcal{O}$
$\boldsymbol{\delta}$	Projection deviation
$\mathbf{x}^*$	Cost-optimal design (original decision-vector space)
$\mathbf{z}^*$	Cost-optimal design in exploratory coordinates
$\mathbf{z}_\mathcal{O}^*, \mathbf{z}_f^*$	Trial point; its projection onto the near-optimal region

Scalars

C, C^*	Total system cost; its optimum
ε	Near-optimality cost slack
$z_i, \bar{z}_i, \tilde{z}_i$	Axis value; its near-optimal maximum; normalised value
$\tilde{z}_C$	Normalised cost coordinate
$d_{\mathcal{I}\mathcal{O}}$	Max–min distance between $\mathcal{I}$ and $\mathcal{O}$ (convergence metric)
n_z, n_x, m	Number of exploratory coordinates; of model variables; of certified designs
$\Delta \text{cap}_{j,n}$	Capacity addition of technology j at node n
$u_{j,n,t}, \tau_t$	Import flow of carrier j ; duration of representative time step t
r_{ij}	Pearson correlation of axes i and j
$\hat{p}$	Rejection-sampling acceptance rate (volume-ratio estimate)
$\hat{R}$	Rank-normalised split convergence statistic

1 Introduction

Mitigating climate change requires fundamental changes in energy systems all around the world [1]. Planning these changes is a complex, long-term task, and policymakers increasingly rely on energy-system optimisation models to support it [2, 3]. From inputs such as technology costs, energy demands and resource potentials, these models compute a design for a future energy system, typically the least-cost one.

Any such model is a simplification of reality, and its results are uncertain. This **uncertainty** is commonly divided into two kinds, **parametric and structural** [4, 5, 6]. **Parametric** uncertainty concerns the **model's input values**, such as future costs and demands, which can only be estimated; it is usually probed by varying these values across **scenarios** [4, 7]. **Structural** uncertainty concerns the **model's own equations**, which **capture a real system only in part**. Much of what drives real decisions never enters them: how readily the **public accepts** new infrastructure, how **politically feasible** it is, or how it competes for land [4, 7, 8].

Because these factors stay outside the model, its **least-cost design** may be neither what is finally built nor what would be chosen in practice; real-world transitions tend to drift from the cost-optimal path [9]. A single optimum, presented on its own, can therefore **mislead**: it **suggests more certainty than the model supports**, and obscures the many alternatives that cost only a little more [7].

One response is to **add the missing factors** to the model as **further objectives**. But each has to be identified and quantified beforehand, and the result is still a **set of optimal trade-offs rather than the wider range of acceptable designs** [4]. Modelling to Generate Alternatives (MGA) takes a different route [4, 10]. Holding the model's inputs fixed, it **maps the near-optimal space**, the designs whose cost lies within a small margin of the optimum. MGA thus **addresses structural rather than parametric uncertainty**, and the factors the model leaves out are for decision-makers to weigh against the alternatives it reveals [7, 11].

What most **MGA methods cannot** do is measure how much of the near-optimal space they have missed. Each of their solves returns a single design on the boundary of the space, and no finite collection of them shows how much is left **unexplored** [12]. Methods that try to capture the **whole space** rather than sample its boundary have so far not scaled past about ten **dimensions** [13]. ORACLE (Optimization for the Rigorous Analysis of the

Complete Landscape of alternatives in Energy systems) closes this gap [14]. It brackets the near-optimal space between an inner and an outer approximation, and reads the distance between them as a certified bound on how much of the space is still unaccounted for.

This thesis **applies ORACLE** to a large, sector-coupled model of the European energy system, exploring a selected set of key decisions. ORACLE was introduced on a smaller, single-node model in six dimensions [14]. The model here is larger, and its **near-optimal space spans ten dimensions**, which is high for studies of this kind [13, 15]. Whether ORACLE still converges in this harder setting is the first research question: *Does ORACLE converge on a large, sector-coupled European energy-system model, and what limits its convergence in practice?*

The bracketed **space** must still be **interpreted**, and drawing insight from it is a challenge in its own right [11]. The **near-optimal designs are many**, and studies usually report only a few, drawn from the **boundary rather than the continuous interior** that contains most of them. **Describing it fully**, at the boundary and in the interior, is the second research question: *What structure does the near-optimal design space have, and which capacity ranges, substitutions and co-requirements hold within it?*

The thesis continues in Section 2 with the **near-optimal space**, the **MGA** methods that explore it, and how the **literature interprets** their results. Section 3 then presents the **modelling framework and energy-system model**, the **ORACLE** method and its application, and the **sampling and interpretation methods**. Section 4 reports the **cost-optimal baseline**, the course of the **exploration**, and the **structure** of the **near-optimal space**. Section 5 **interprets** these results, for ORACLE as a method and for the energy system, and sets out the **limitations**. Section 6 answers the research questions and closes with an **outlook**.

2 Background

This chapter introduces the concepts this work builds on. Section 2.1 defines the near-optimal space, presents the MGA methods that explore it, and reviews their limits. Section 2.2 turns to what the literature learns from the explored space, and distinguishes its boundary from its interior, the distinction that frames the interpretation in this work. The modelling framework, the energy-system model and the ORACLE method itself follow in the Methodology (Section 3).

2.1 Modelling to Generate Alternatives and the near-optimal space

Energy-system optimisation models compute the design of an energy system that is optimal under a chosen objective, most commonly the total system cost. Minimising the total cost over all feasible designs is a linear program (LP), whose optimal value is the cost optimum

$$C^* = \min_{\mathbf{x}} \mathbf{c}^\top \mathbf{x} \quad \text{s.t.} \quad \mathbf{x} \in \mathcal{X}, \quad (1)$$

where $\mathcal{X}$ is the feasible space, the set of all designs $\mathbf{x}$ that satisfy the model's constraints, $\mathbf{c}^\top \mathbf{x}$ the total cost of a design, and $\mathbf{x}^*$ the cost-optimal design attaining C^* .

MGA widens the view from this single optimum to the near-optimal space: the feasible designs whose cost exceeds the optimum by no more than a chosen fraction, the cost slack $\varepsilon \geq 0$ [4, 7, 9]. These designs form the near-optimal feasible space,

$$\mathcal{X}_\varepsilon = \{\mathbf{x} \in \mathcal{X} \mid \mathbf{c}^\top \mathbf{x} \leq (1 + \varepsilon) C^*\}. \quad (2)$$

Because the model is an LP, $\mathcal{X}_\varepsilon$ is a convex polytope: the feasible space intersected with the cost half-space (Figure 1).

MGA maps this space by re-solving the model with the cost objective replaced by an alternative one, while cost is kept as a constraint,

$$\min_{\mathbf{x}} \mathbf{w}^\top \mathbf{x} \quad \text{s.t.} \quad \mathbf{x} \in \mathcal{X}, \quad \mathbf{c}^\top \mathbf{x} \leq (1 + \varepsilon) C^*, \quad (3)$$

whose feasible set is exactly the near-optimal space $\mathcal{X}_\varepsilon$ of Eq. (2). The weight vector $\mathbf{w}$

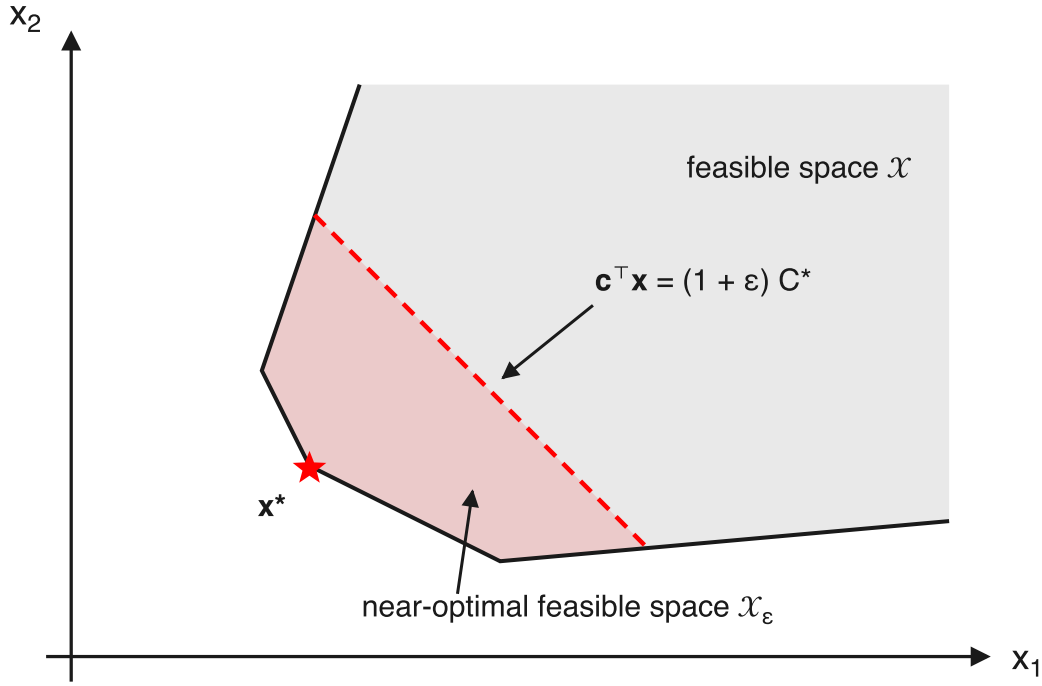


Figure 1: Two-dimensional schematic of the near-optimal space. The grey polytope is the feasible space $\mathcal{X}$, and its vertex $\mathbf{x}^*$ the cost optimum. The dashed line is the cost bound $\mathbf{c}^\top \mathbf{x} = (1 + \varepsilon) C^*$; the feasible designs on its optimal side (red) make up the near-optimal space $\mathcal{X}_\varepsilon$ of Eq. (2).

prices the decision variables and thereby steers each solve towards a different part of the space [4, 10]. Because the **objective stays linear, each solve is again an LP**, and the **design** it returns **lies on the boundary of $\mathcal{X}_\varepsilon$** , generically at a **vertex**.

The **MGA methods** in the **literature differ** mainly in one choice: **how they set the weights** [12, 14]. **Hop-Skip-Jump (HSJ)**, the earliest method, **builds each new objective from the previous solutions**: every decision variable is weighted by how many of them used it, and minimising this weighted sum **drives the next design** away from the **technologies already built** out [4, 10]. Because each solve depends on all previous ones, the **method cannot run in parallel**, and it tends to stay **close to the cost-optimal design** rather than cover the space [12]. **Variable min/max (VMM)** instead picks a **set of decision variables in advance** and **drives each to its minimum and its maximum** in turn, which pins down their exact extremes but returns only **two designs per variable** and nothing between them [7, 16, 17]. **Random objective vectors**, in contrast, draw the **weights at random** and so **spread** the search **evenly, reaching every vertex** of the near-optimal space in the limit of **unlimited iterations**, but they **rarely point straight** along a single axis and therefore **miss its extremes** [12]. These are not all of the methods, but they illustrate how the choice of weights trades off coverage against extremes.

The designs found by any of these methods implicitly **span a region**, their **convex hull**.

Modelling All Alternatives (MAA), a **region-based** MGA method, makes this region the engine of the search [13]. Starting from VMM extremes, it builds the convex hull of all designs found so far, **re-solves** along the **outward normals of the hull’s facets**, and **repeats until the hull volume stops growing** appreciably. The hull must be constructed explicitly for this, with the **Quickhull algorithm** [18], and this construction limits the method: the **effort** of building the hull grows steeply with the **number of explored dimensions** [14, 18], and it proved **infeasible beyond ten dimensions** even on large hardware [12, 13]. The **volume-based stopping rule** is moreover a **heuristic**, and MAA converges to the near-optimal space only in the limit of unlimited iterations [14]. A later variant replaces the heuristic with a valid **volume-based convergence metric** [19], but keeps the explicit hull and with it the dimension limit.

One limitation is common to all of these methods: the exploration ends after a set number of solves, or when a heuristic stops improving, and neither tells **how much** of the near-optimal space remains uncharted [12, 14]. A stagnating hull volume, in particular, may only mean that the method has found all the designs it is able to find, not that the **space is covered** [12, 14]. ORACLE closes this gap by additionally bounding the near-optimal space from the outside, which turns the **distance** between the two bounds into a certified **error measure** [14]. ORACLE is the exploration method of this work; the Methodology presents it in detail (Section 3.2).

2.2 Interpreting high-dimensional design spaces

An MGA search produces a set of near-optimal designs, which most studies interpret directly [11]. They present a **small number of maximally different alternatives** [4, 12], or they aggregate the set into the **ranges technologies can span** [7], **how frequently** technologies appear across the alternatives [9, 20], and the **spatial patterns** the alternatives form [20, 21]. Ranges, however, are easily over-read: they suggest that their ends are simultaneously achievable, while in reality **fixing one technology near its extreme restricts the values available to every other** [11]. Nor does the **frequency** with which a value appears across the alternatives carry **practical significance for decision support** [11].

Other studies ask **not what is possible** but **what is unavoidable**. Necessary conditions are **minimum requirements** that every design within the cost bound must meet, for example a minimum total capacity of a technology group [7, 22]. Statements of this kind hold

for the **entire near-optimal space**, independent of which alternatives a particular search happens to find.

None of these readings **describes the interior of the space**. That interior is reached through the **convex hull** that the explored designs span (Section 2.1): because any **mixture of near-optimal designs is again near-optimal**, **every point** of the hull is itself a usable near-optimal design [14]. Studies have **sampled** this hull uniformly to see how the **near-optimal designs are distributed** [13], taken points **deep inside** it as **candidate robust designs** [15], and **generated further designs** from it without re-solving the model [11]. Such interior readings **describe volume, not likelihood**: a design that lies deeper inside the space is not thereby more probable than one near the boundary [11].

Whatever is read from the interior is read from this hull, and the **hull is only an inner approximation**: none of the methods above can say how much of the near-optimal space lies outside it. This is where ORACLE differs [14]. It bounds the space with an outer approximation as well, so together with the inner hull the near-optimal space is enclosed from both sides, and the error of reading it off either body becomes certified. It **keeps the hull implicit**, testing membership with an LP, so the Quickhull limitation of Section 2.1 never arises. Its **convergence metric is moreover method-agnostic**, able to certify the coverage of any MGA-generated design set.

This work draws on both the **boundary and the interior of the near-optimal space**. The **boundary** sets the **outer limits**: how far each technology can be pushed, and which investments no near-optimal design can avoid. The interior describes what lies within them: **where most of the near-optimal volume sits**, and which **technologies rise and fall together**. This thesis derives **certified ranges and necessary conditions** and characterises the interior; portfolios of **deliberately extreme designs [21] are outside its scope**. The methods that interpret the explored space are selected in the Methodology (Section 3.4).

3 Methodology

This chapter presents the methodology of this work. Section 3.1 introduces the **modelling framework and the energy-system model** whose near-optimal space we explore. Section 3.2 presents the **ORACLE method**, our choice of **exploratory variables**, the **application** of the method to the model, and the treatment of **cost as an additional exploratory variable**. Section 3.3 describes how **uniform samples are drawn from the resulting approximations**, and Section 3.4 the **analyses** used to **interpret** them. Figure 2 gives a graphical overview.

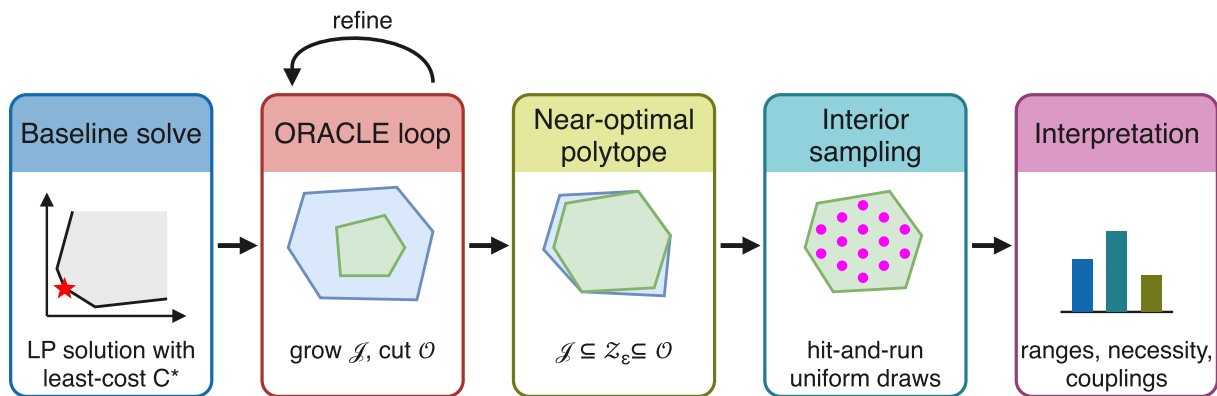


Figure 2: The end-to-end workflow, from the cost-optimal baseline to the interpreted near-optimal space. A **least-cost solve** of the model returns the cost optimum C^* and the cost-optimal design. From these, the **ORACLE loop** refines an inner approximation $\mathcal{I}$ (green) and an outer approximation $\mathcal{O}$ (blue) that bracket the near-optimal space, until the **gap** between the two falls **below a tolerance**. The **converged bracket** is then **sampled uniformly** in its interior by **hit-and-run**, and the resulting **draws (magenta)** are **interpreted** as capacity ranges, necessary conditions and couplings between technologies.

3.1 ZEN-garden and the Crystal Ball model

Framework and dataset play different roles in this work: ZEN-garden is the mathematical framework that builds and solves the optimisation model, and the Crystal Ball dataset defines the energy system it describes, our case study. We introduce both in turn.

ZEN-garden

ZEN-garden is an open-source framework for the optimisation of energy-system transition pathways [23]. From a dataset that defines technologies, carriers, nodes and their parameters, the **framework builds an LP**, typically spanning a multi-year horizon that is solved either with perfect foresight or myopically with a rolling horizon. The available objective functions are the **minimisation** of the **cumulative net present cost** or of the cumulative carbon emissions of the energy system over the planning horizon; this work minimises

cost. The **decision variables** cover **investment and operation**: the **capacity additions** per technology, location and year, and the operational variables, namely the **conversion, storage and transport flows, storage levels, imports and exports**. The **constraints** enforce nodal **energy balances** for every carrier and time step, **capacity limits and potentials, storage dynamics**, and the **carbon policy**, here annual emission limits and a cumulative budget, each with a priced overshoot. ZEN-garden solves its models with the open-source solver HiGHS by default; we use **Gurobi** instead, with the settings listed in the Appendix (Table A2).

The **greenfield-2050 Crystal Ball** model

The Crystal Ball dataset defines the **sector-coupled European energy-system model** of this work [24]. The dataset comprises **28 nodes, 31 carriers and 84 technologies**, of which 68 are conversion, 6 carbon capture and storage (CCS) retrofit, 5 storage and 5 transport technologies (Appendix, Figure A1). Crystal Ball was built to **optimise the European energy transition from today to 2050 as a pathway under perfect foresight** [24]. We build on a slightly **modified** version of the dataset.

Two modifications turn the pathway dataset into our **greenfield-2050 configuration**. First, we reduce the horizon to the **single target year 2050**: instead of the transition towards it, the model **designs the 2050 system directly**. Studying a single target year in this way is **common in the near-optimal literature** [7, 13, 15], and it separates the question of **what the 2050 system should look like** from the question of how to get there. Second, we **disable** the **technology-diffusion limit**, which caps annual capacity additions as a fraction of the previously built stock. This limit is **calibrated for a gradual pathway** and would forbid building the 2050 system within a single year.

The configuration is **not a strict greenfield**. The model **inherits the historical capacity stock**, but technology lifetimes retire nearly all inherited conversion capacity before 2050. What substantially survives is **long-lived network and storage infrastructure, such as gas pipelines, power lines, gas storage and parts of the hydropower fleet**. Since our exploratory variables measure what is newly built or imported for 2050 (Section 3.2), the inherited stock only shapes the background system.

Demand and technology data follow the 2050 values of the Crystal Ball specification; technology **costs** are **exogenous, year-indexed inputs** without endogenous learning [24]. The

dataset also carries **exogenous limits**, among them **capacity caps** for individual technologies and **import availabilities** for individual carriers; those **affecting the explored quantities** are listed in the Appendix (Table A6). **Operations** are resolved on **10 system-wide representative time steps per year**, obtained by time-series aggregation (**TSA**). The Appendix summarises the full configuration (Table A1). The **cost optimum C^*** of this configuration anchors the near-optimality constraint throughout this work.

3.2 ORACLE: the metric and method

MGA methods explore the near-optimal space through the designs they generate, but they **cannot state how much of the space those designs miss** (Section 2.1). ORACLE closes this gap with **two methodological contributions** [14]: a **metric** that measures how much of the space is still uncharacterised, and a **method** that **drives this metric below a tolerance**. This section presents both in their general form; the following sections describe our **choice of exploratory variables** and the **application** to the greenfield-2050 model.

Exploring the **near-optimal design space $\mathcal{X}_\epsilon$** of Eq. (2) **directly** is intractable, because a full design $\mathbf{x}$ of a real energy-system model has **hundreds of thousands of variables**, while the **decisions of interest are a handful of aggregate quantities**. ORACLE therefore **explores** the **space of exploratory variables $\mathbf{z} = \mathbf{S}\mathbf{x}$** , where $\mathbf{S} \in \mathbb{R}^{n_z \times n_x}$ is a **linear map** that condenses the **n_x model variables** into few **interpretable quantities**. In these variables, the **near-optimal** space becomes

$$\mathcal{Z}_\epsilon = \{\mathbf{z} \mid \mathbf{z} = \mathbf{S}\mathbf{x}, \mathbf{x} \in \mathcal{X}_\epsilon\}, \quad (4)$$

the **image of $\mathcal{X}_\epsilon$ under $\mathbf{S}$** ; notation follows [14].

ORACLE brackets the near-optimal space between two approximations. **While exploring**, the **method collects certified near-optimal designs**, and the **inner approximation $\mathcal{I} \subseteq \mathcal{Z}_\epsilon$** is their **convex hull**, kept *implicitly* as

$$\mathcal{I} = \{\mathbf{z} \mid \mathbf{z} = \mathbf{Z}^\top \boldsymbol{\lambda}, \mathbf{1}^\top \boldsymbol{\lambda} = 1, \boldsymbol{\lambda} \geq 0\}, \quad (5)$$

where $\mathbf{Z} \in \mathbb{R}^{m \times n_z}$ holds the **m designs as rows** and **$\boldsymbol{\lambda}$ are convex-combination weights** [14]. The hull is **never constructed explicitly**: whether a **point lies in $\mathcal{I}$** can be **checked** by a **small LP**. The **outer approximation $\mathcal{O} \supset \mathcal{Z}_\epsilon$** **encloses** the space in an **intersection** of

half-spaces,

$$\mathcal{O} = \{ \mathbf{z} \mid \mathbf{A}\mathbf{z} \leq \mathbf{b} \}, \quad (6)$$

each of which is known to contain the whole near-optimal space; the exploration adds half-spaces as it learns about the space, tightening $\mathcal{O}$ from the outside.

The method refines this bracket in five steps, shown in Figure 3 [14].

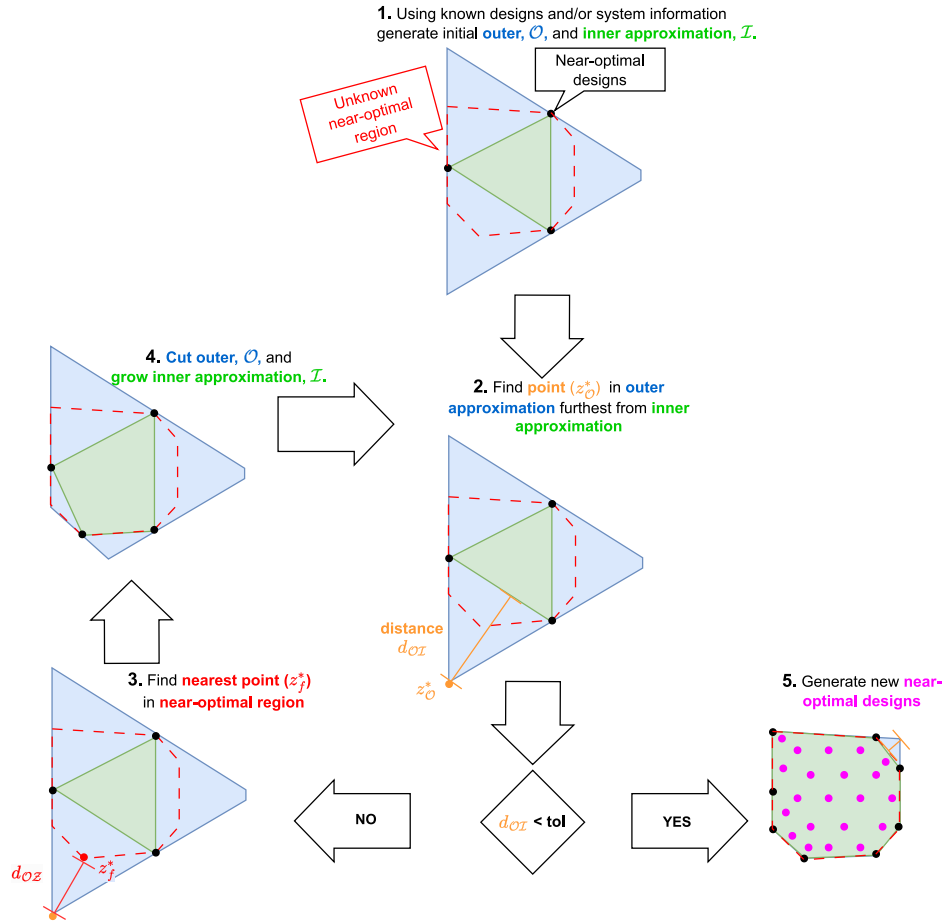


Figure 3: The five steps of the ORACLE algorithm. From an initial inner/outer bracket (Step 1), each iteration finds the outer point z_O^* farthest from the inner approximation (Step 2), projects it onto the near-optimal region to find the nearest near-optimal design z_I^* (Step 3), and uses that design to grow the inner approximation and cut the outer one (Step 4). The loop repeats until the max-min distance d_{IO} falls below the tolerance, when near-optimal designs are generated from the converged bracket (Step 5). Reproduced from [14].

Step 1 (initialise): from known near-optimal designs and system information, ORACLE forms the initial inner and outer approximations. The cost-optimal design is always known and seeds the inner approximation, and any further known designs can be added to it; bounds on the exploratory variables give the initial half-spaces of the outer approximation.

Step 2 (trial point): each iteration finds the trial point z_O^* , the point of the outer

approximation that lies farthest from the inner one, by solving the bilevel problem

$$d_{\mathcal{IO}} = \max_{\mathbf{z}_O \in \mathcal{O}} \min_{\mathbf{z}_I \in \mathcal{I}} \|\mathbf{z}_O - \mathbf{z}_I\|_\infty, \quad (7)$$

an optimisation over the two approximations alone, with no model solve involved. Because $\mathcal{I} \subseteq \mathcal{Z}_\varepsilon \subset \mathcal{O}$ holds throughout, the optimal value $d_{\mathcal{IO}}$ is an upper bound on the error of reading the near-optimal space off either approximation. The same quantity therefore steers the search and measures its convergence. If it is below the chosen tolerance, the loop proceeds to Step 5.

Step 3 (project): the trial point is projected onto the near-optimal space by solving the full energy-system model once, with a distance-minimising objective in place of the weighted-MGA objective,

$$\min_{\boldsymbol{\delta}, \mathbf{z}_f, \mathbf{x}} \|\boldsymbol{\delta}\|_\infty \quad \text{s.t.} \quad \mathbf{z}_O^* - \mathbf{z}_f = \boldsymbol{\delta}, \quad \mathbf{z}_f = \mathbf{S}\mathbf{x}, \quad \mathbf{x} \in \mathcal{X}_\varepsilon, \quad (8)$$

where $\boldsymbol{\delta}$ is the distance between the trial point and $\mathbf{z}_f$, the feasible image point the projection searches for, and $\mathbf{x}$ the full design realising it. The solution $\mathbf{z}_f^*$ is the point of the near-optimal space nearest to the trial point.

Step 4 (refine): the projection result refines both approximations. The point $\mathbf{z}_f^*$ is added to the certified designs in $\mathbf{Z}$, growing the inner approximation. If the trial point lay outside the near-optimal space, a new inequality that separates it from that space is added to $\mathbf{A}\mathbf{z} \leq \mathbf{b}$, shrinking the outer approximation; if the trial point was already near-optimal, no cut is possible and only the inner approximation grows. The loop returns to Step 2.

Step 5 (terminate): once $d_{\mathcal{IO}}$ falls below the tolerance, the two approximations are accepted. Near-optimal designs and further analyses can then be generated from them at negligible computational cost, without further model solves.

Solving the max–min problem (7) is a challenge in itself. The problem is bilevel, because the maximisation over $\mathcal{O}$ contains the minimisation over $\mathcal{I}$. With the infinity norm, both levels are linear, and the bilevel problem can be reformulated into a single-level problem [14]; the reformulation used here is discussed with the application below. ORACLE is the only MGA-family method that guarantees convergence to within a chosen tolerance [14]. The guarantee rests on a premise the paper leaves implicit: every max–min solve must

be closed to proven optimality, or nearly so, because any gap a solve leaves open carries into the certificate and widens the tolerance that can actually be guaranteed.

Exploratory variables

Choosing the exploratory variables $\mathbf{z}$ means choosing the linear map $\mathbf{S}$, and the map has much to condense. For every conversion, storage and transport technology at every node, the model holds capacity-addition variables, and for every representative time step it additionally holds the operational variables, from conversion flows to storage levels and imports. In the greenfield-2050 configuration, this adds up to $n_x = 467\,280$ variables. Each exploratory variable aggregates one group of these variables into a single quantity. The exploratory variables are the coordinates of the explored space, so we also call them axes; each axis is one row of $\mathbf{S}$, and $i = 1, \dots, n_z$ indexes them.

A technology axis sums the newly built 2050 capacity of its member technologies over all locations,

$$z_i = \sum_{j \in \mathcal{M}_i} \sum_{n \in \mathcal{N}} \Delta \text{cap}_{j,n}, \quad (9)$$

where the member set $\mathcal{M}_i$ contains the technologies aggregated by axis i , $\mathcal{N}$ is the node set and $\Delta \text{cap}_{j,n}$ the capacity addition of technology j at node n . For storage technologies, only the energy capacity enters, not the power capacity. A carrier-import axis is the annual duration-weighted import of its member carriers,

$$z_i = \sum_{j \in \mathcal{M}_i} \sum_{n \in \mathcal{N}} \sum_{t \in \mathcal{T}} \tau_t u_{j,n,t}, \quad (10)$$

with $u_{j,n,t}$ the import flow of carrier j and τ_t the duration of representative time step t . Building exploratory variables as such sums follows established MGA practice [13]. We explore ten variables. Eight are technology axes, one for each of nuclear, photovoltaics (PV), offshore wind, onshore wind, electrolysis, direct air capture (DAC), battery storage and an aggregate of CCS technologies. The ninth axis is the biomass import, and the tenth is the total system cost, whose special role is explained below. These technologies and the biomass resource are among the most debated design choices of the European energy transition [1, 24]. The CCS lump aggregates all six capture retrofits of the model into one axis, to keep the number of explored dimensions small. Its members are the capture retrofits at blast-furnace steel plants, biomass plants, cement kilns, gas turbines,

gas-based direct-reduction steel plants and steam-methane reformers; the exact **dataset identifiers** of all axes are **listed** in the **Appendix** (Table A6).

The axes **measure different quantities in different units** (GW, GWh, ktCO₂eq/h, MEUR). Under the L_∞ **metric** of Eq. (7), axes with **numerically large ranges** would dominate both the **exploration and the polytope geometry**. We therefore **normalise** the nine design axes by their **near-optimal maxima**,

$$\tilde{z}_i = \frac{z_i}{\bar{z}_i}, \quad \bar{z}_i = \max \{z_i \mid \mathbf{z} \in \mathcal{Z}_\varepsilon\}, \quad (11)$$

so that each of them ranges from 0 to 1 within the **near-optimal space**; the cost axis is normalised below.

Applying ORACLE to the energy-system model

Two pieces of software **connect ORACLE to the greenfield-2050 model**. The loop runs in the **reference implementation of the ORACLE authors (pyoNearOpt)** [14], extended for this work with the **strong-duality certificate** introduced below. A **ZEN-garden plugin** developed for this work adds the **near-optimality constraint to the model, swaps objectives between solves, and answers each projection query** (Step 3, Eq. (8)) with one **solve of the full model, in the normalised coordinates**.

For the **initial outer approximation**, the reference implementation uses the **variable bounds of the model** [14]. Our axes are aggregates for which the dataset provides **no such ready-made finite bounds**, because **capacity additions are unbounded by default**. We therefore **compute the near-optimal maximum $\bar{z}_i$ of every axis**, as defined in Eq. (11), with **one LP over the full model**. In **normalised coordinates**, the initial outer approximation is then the **unit box $0 \leq \tilde{\mathbf{z}} \leq 1$** , and the maxima simultaneously serve as the **normalisation constants**. The **inner approximation starts** from the **cost-optimal design alone**.

The **cutting planes** of Step 4 are **built from duals of the projection solves**. We solve the model with the **barrier method without crossover** (Table A2), whose **duals** are **exact** only up to the **solver's convergence tolerance**, so a **cut** can **accidentally slice** into the **near-optimal space**. Every new cut is therefore **checked against all certified points** and **relaxed** if it would cut one off; this **preserves the inclusion $\mathcal{O} \supseteq \mathcal{Z}_\varepsilon$** , on which the error bound relies.

We explore with a **cost slack of $\varepsilon = 0.1$** , i.e. designs within 10% of C^* , and target a

convergence tolerance of 0.1 in normalised units. The full run configuration is listed in the Appendix (Table A3).

Certifying convergence rests on solving the max-min problem (7), which is the hardest part of the loop in practice. The problem is a bilevel program; bilevel programs are NP-hard and cannot be handed to a standard solver directly [14, 25], so the problem must first be reformulated into a single-level one. The reference implementation reformulates it through the Karush–Kuhn–Tucker (KKT) optimality conditions of the inner minimisation, which turns the bilevel program into a single-level mixed-integer linear program (MILP) that a general-purpose solver can process (Appendix A.2 of [14]), enforcing the complementarity conditions through big-M constants [26]. On our refined approximations, this MILP failed to close its bound within the time budget. We therefore reformulated the same max-min problem through strong duality instead [27, 28], which yields a bilinear program without integer variables, a non-convex quadratic program (QP) that a global solver drives to tighter bounds in the same time. Because the loop needs the trial point rather than the proof, each in-loop solve runs under a short time limit, and the proof is bought once, by a single long certificate solve on the final approximations (Appendix, Table A3). This has one consequence for the stopping rule: ORACLE stops once the proven bound falls below the tolerance, but the bounds of our time-capped in-loop solves stall well above it, so the loop would not terminate on its own. We therefore fixed the iteration budget in advance, at 700 iterations, estimated from the incumbent trajectories of earlier runs as the point where the true distance should approach the tolerance.

A separate correction concerns how the bound is read, not how it is computed. When the KKT complementarity conditions are instead encoded with special ordered set of type 1 (SOS1) constraints, the reformulated model carries no integer variables, and the optimisation-modelling library Pyomo then returns the incumbent objective in place of the solver’s dual bound. Only the dual bound is a valid upper bound on $d_{\mathcal{IO}}$, so we read it straight from the solver: this changes nothing under the big-M encoding, which the library reads correctly, and corrects for the SOS1 one. Whenever a certificate solve reaches its time limit before closing its gap, we report the metric as a bracket on $d_{\mathcal{IO}}$, between the incumbent as a valid lower bound and the dual bound as a valid upper bound (Section 4.2).

Cost as an exploratory variable

In **MGA**, the **total system cost** moves from the objective into the **constraints** (Section 2.1): cost is no longer optimised, only bounded from above, and every design may move freely within the cost band $[C^*, (1 + \varepsilon) \cdot C^*]$. Within this band, the **solver is indifferent to cost**, and the allowed cost increase, the slack $\varepsilon \cdot C^*$, is spent on whatever incidentally **helps the current exploration objective**; in a full-pathway test run, the slack **leaked** into the **carbon-overshoot penalty**. Such behaviour is **invisible** in the design axes: two designs with identical capacities, one at C^* and one at $1.1 \cdot C^*$, are the same point of the design space.

To **make the cost level observable**, we append the cost itself as the tenth exploratory variable: the net present cost $C = \mathbf{c}^\top \mathbf{x}$ that the model otherwise minimises, **normalised** over the allowed cost band,

$$\tilde{z}_C = \frac{C - C^*}{\varepsilon \cdot C^*}. \quad (12)$$

The explored vector therefore becomes $\tilde{\mathbf{z}} = (\tilde{z}_1, \dots, \tilde{z}_9, \tilde{z}_C) \in [0, 1]^{10}$. Formally, cost is simply one more linear aggregate of the design vector, with the cost vector $\mathbf{c}^\top$ as its row of $\mathbf{S}$, so the **explored set remains an image of $\mathcal{X}_\varepsilon$** exactly as in Eq. (4). In the language of **weighted MGA**, exploring this vector amounts to allowing a **weight on total cost** while the **near-optimality constraint stays in place**: the exploration may pull cost up or down within the band, but never beyond it.

The cost axis **changes what the exploration sees**. Each trial point now prescribes a **cost level** alongside the **design targets**, and the projection treats deviations from it like deviations on any other axis, so the **cost of every certified design** is **controlled by the exploration** instead of falling out of the solver arbitrarily. The **slack** turns from solver noise into an **explored quantity**: the same capacity mix at different cost levels becomes distinct points of the explored body, and for every mix the body records the range of costs at which it is achievable. The **axis** deliberately does **not steer the exploration towards cheap designs**. A fixed cost weight would do that, but it would **map only the cheap edge of the band**, and a **complete map** of the near-optimal space must include the expensive part as well.

The cost coordinate has one **consequence** for all later **statistics**. In the ten-dimensional body, **each capacity mix** extends along the cost axis across the **whole range of cost levels** at which it is achievable. A **mix achievable across a wide cost range** thus occupies **more vol-**

une than one confined to a narrow range, and a uniform sample over the ten-dimensional body lands on it correspondingly more often: projected onto the nine design axes, such a sample would over-represent the cost-flexible mixes. All design-space statistics in this work are therefore computed on nine-dimensional bodies from which the cost axis has been eliminated exactly, before any sampling takes place (Section 3.3). The ten-dimensional body remains available for cost-conditional analyses.

3.3 From the polytope to a sample set

Characterising the interior of the near-optimal space requires points from that interior, and optimisation cannot provide them, because LP solves return boundary points. We therefore turn to uniform sampling. The near-optimal space is convex and hence connected, so a random walk can cover it; what remains to decide is which of the two bodies to walk on.

Sampling operates on the nine-dimensional design bodies motivated in Section 3.2, so the cost axis is removed exactly before any draw. For the inner body, each certified design is a row of $\mathbf{Z}$, and the cost values of all designs form one column; deleting this column projects every design onto the design axes, and the convex hull of the projected designs is exactly the projection of the hull. For the outer body, the cost variable is eliminated from the half-space description by Fourier–Motzkin elimination [29], which combines pairs of inequalities such that the cost variable cancels, followed by an LP-based removal of redundant rows.

Uniform interior samples are rare in the near-optimal literature. The original MAA study drew them by decomposing its explicit convex hull into simplices and sampling each simplex in proportion to its volume [13]. This route requires the explicit hull, and constructing one from several hundred certified points in nine dimensions is precisely the Quickhull limitation described in Section 2.1. We therefore sample with a Markov-chain Monte Carlo (MCMC) method and rejection instead.

The walk we use is hit-and-run [30, 31]: from the current point, draw a random direction, then move to a uniformly chosen point on the chord through the body. To cross elongated bodies efficiently, we precondition the directions with the covariance of the certified points. The two bodies differ in what a walk on them costs and what they guarantee. The outer body is described by plain half-spaces, so walking on it is cheap. The inner body is

reachable only through the **convex-combination weights** of Eq. (5), so **every step** of a walk on it **costs two LPs** over these weights, one per chord endpoint. In exchange, every point of the **inner** body is a **certified near-optimal design**, whereas the outer body also contains volume that is not near-optimal. Both bodies **deviate** from the true region by at most the final **max-min distance**: the inner body under-covers it, the **outer body over-covers** it. Because we want samples that are certified near-optimal, we **sample the inner body**. **Two routes** lead to samples of the inner body. The **direct route** walks on the inner body itself and **pays the two chord LPs** at every step; **consecutive steps** are moreover **strongly correlated**, so **many steps are needed** for every effectively independent draw. The **rejection route** walks on the **outer body** instead and **pays one membership LP per proposal**, keeping exactly the draws that land in the inner body. Restricting a uniform sample to a subset leaves it exactly uniform on that subset [30], and the **acceptance rate**

$$\hat{p} = \frac{n_{\text{accepted}}}{n_{\text{proposals}}} \approx \frac{\text{vol}(\mathcal{I})}{\text{vol}(\mathcal{O})} \quad (13)$$

estimates the **volume ratio of the two projected bodies**, where $n_{\text{proposals}}$ counts the uniform draws from the outer body and n_{accepted} those passing the membership test. **Which route is faster depends on this acceptance rate**; in our setting the rejection route reached a given **effective sample size (ESS)** several times faster, so we use it as the primary sampler.

To validate this shortcut, we ran the direct walk once as an independent cross-check and compared the two sample sets. **Per axis**, the samples are compared with the **Kolmogorov-Smirnov test**, which measures the **largest difference between their cumulative distributions**. Per axis pair, the draws are binned on a grid and the bin counts are compared with a **chi-squared test**. Both comparisons are Bonferroni-corrected for the many tests performed; their outcome is reported with the **results** (Section 4.3).

Random-walk samples are correlated, and a single walk can appear settled while it has **explored only part of the body**. Judging convergence therefore requires comparing **several independent walks**; we run eight in parallel from dispersed starting points and accept the sample set only if the **walks are statistically indistinguishable**. The rank-normalised **split- $\hat{R}$ statistic** [32] compares the **spread between walks with the spread within** them and **approaches 1 as they mix**; we require $\hat{R} < 1.01$ on every axis. The **ESS** discounts the correlation within the walks and states **how many independent draws** the sample set is worth [32]; it is gated against a target. Beforehand, a pilot phase measures the walk's

correlation and acceptance and sizes the burn-in and the proposal budget for the targeted ESS. The **sampler is implemented in Python** for this work, with the **membership and chord LPs solved by Gurobi** and the **diagnostics computed with ArviZ**; the full configuration is listed in the Appendix (Table A5).

A sample is a **design vector $\mathbf{z}$ only, with no full design $\mathbf{x}$ behind it**. The **cost axis** is therefore the only way to know a **sampled design's cost** (Section 3.2). If a **full design** is needed, **any sample can be realised by one projection solve**, Eq. (8), because every point of the inner body is the image of at least one near-optimal design.

3.4 Interpretation methods

Three analyses interpret the near-optimal space: **certified necessary conditions**, the **distribution** of the near-optimal volume and **pairwise correlations**. The **necessary conditions** are computed on the **full model** and are **independent of the approximations and the sampling**; the **two other analyses** build on the **uniform draws** of Section 3.3.

Necessary conditions

Which **investments are unavoidable** in every near-optimal design? To answer this, we compute for every **axis** the **smallest and the largest value** attainable within the near-optimal space,

$$\min / \max \{z_i \mid \mathbf{z} \in \mathcal{Z}_\varepsilon\}, \quad (14)$$

where each **bound** is one **exact LP over the full model subject to the near-optimality constraint**, with the **axis value as objective**, just as for the maxima of Eq. (11). The **bounds** are therefore **certified statements about the true near-optimal region** rather than about the polytope approximations, and they are **independent of the ORACLE run**. They are also the solves a **VMM** exploration would run (Section 2.1), used here not to generate boundary designs but to **certify the extent of the space**. The cost axis plays no role here beyond the near-optimality constraint itself.

Beyond the **nine axes**, we compute the **same bounds for three aggregates**: the **total capacity** of variable renewable energy (**VRE**), i.e. PV plus onshore and offshore wind; the **total CO₂-capture capacity**, i.e. DAC plus the CCS lump; and the **total CO₂-removal capacity**, i.e. DAC plus the biomass-fed capture member of the lump, the only two routes that draw

carbon from the atmosphere, whereas the remaining members intercept fossil and process emissions at their point sources. Each **aggregate sums members that share one unit**. **Two conditional bounds** complete the set: to **test** whether the **two negative-emission routes can stand in for each other**, we pin the **biomass import to zero** and **minimise DAC**, and **conversely**, each again one exact LP. A **strictly positive minimum** certifies a **technology, aggregate or conditional floor** as necessary in every near-optimal design (cf. [7, 22]).

The distribution of near-optimal designs

Where **within its bounds** does the **near-optimal volume** lie? Ranges alone treat every value within them as equally available, but the **volume of the body is not spread evenly**: knowing where **most near-optimal designs** lie shows how a **choice on one axis changes the room left on the others** [13]. **Computing the volume** of a nine-dimensional polytope exactly is **out of practical reach** [12]. The **uniform draws** provide the same information instead: under the **uniform measure on the inner body**, the **share of draws** that falls into any region **estimates that region's share of the total volume**. The **empirical densities** of the draws are therefore our volume statements, and the acceptance rate of Eq. (13) is one such estimate.

Read this way, the **marginal distribution** of one axis shows **how much volume remains** when that **axis is pinned to a given value**. A **wide, flat** marginal means the choice **barely constrains** the rest of the system; a **peaked** marginal means **most near-optimal designs agree on that value**. **Pairwise densities** extend the same reading to pairs of axes. We display the pairwise densities in a **matrix of all axis pairs**, each panel outlined with the exact two-dimensional projection of the inner body, and the per-axis marginals on the diagonal.

Pearson correlation

Beyond single axes, we ask **how the axes co-vary across the near-optimal volume**. We measure **pairwise dependence by the Pearson (product-moment) correlation r_{ij}** , the standard estimator: the **covariance of axes i and j** over the uniform draws, scaled by the product of their standard deviations, so that the coefficient ranges from **-1 to 1**. A **negative coefficient** means the **volume trades the two axes off against each other (substitution)**; a positive one means they tend to **rise together (co-requirement)**. The coefficient is invariant

under rescaling of the axes, so the normalisation does not affect it.

Pearson **captures only linear dependence**: a coefficient **near zero** does not rule out a nonlinear relationship. We therefore checked the same draws with measures that **detect monotone and general dependence**, namely Spearman rank correlation, distance correlation [33], Kraskov–Stögbauer–Grassberger (KSG) mutual information [34] and a round-robin ordinary-least-squares (OLS) regression. Their outcome is reported with the results (Section 4.3).

4 Results

This chapter reports the results of applying the methodology to the greenfield-2050 model. Section 4.1 presents the cost-optimal baseline and contrasts it with two near-optimal alternatives. Section 4.2 reports how the ORACLE exploration and its convergence certificate behaved, the subject of the first research question. Section 4.3 then characterises the explored space itself, the subject of the second: certified necessary conditions, the distribution of the near-optimal volume, and the correlations between the design axes.

4.1 Cost-optimal baseline and near-optimal alternatives

The cost-optimal solve of the greenfield-2050 model reaches a net present cost of $C^* = 1\,258\,542$ MEUR; this value anchors the near-optimality constraint of all following results. Measured on the nine design axes, the baseline builds a system carried by onshore wind (1554 GW) and PV (2688 GW), complemented by 286 GW of electrolysis, 65 GW of offshore wind, 24 GW of nuclear, 745 GWh of battery storage and 62 ktCO₂eq/h of capture capacity in the CCS lump. It builds no DAC at all, and it imports biomass at the dataset’s full availability limit of 778 TWh [24], the only axis at its upper bound.

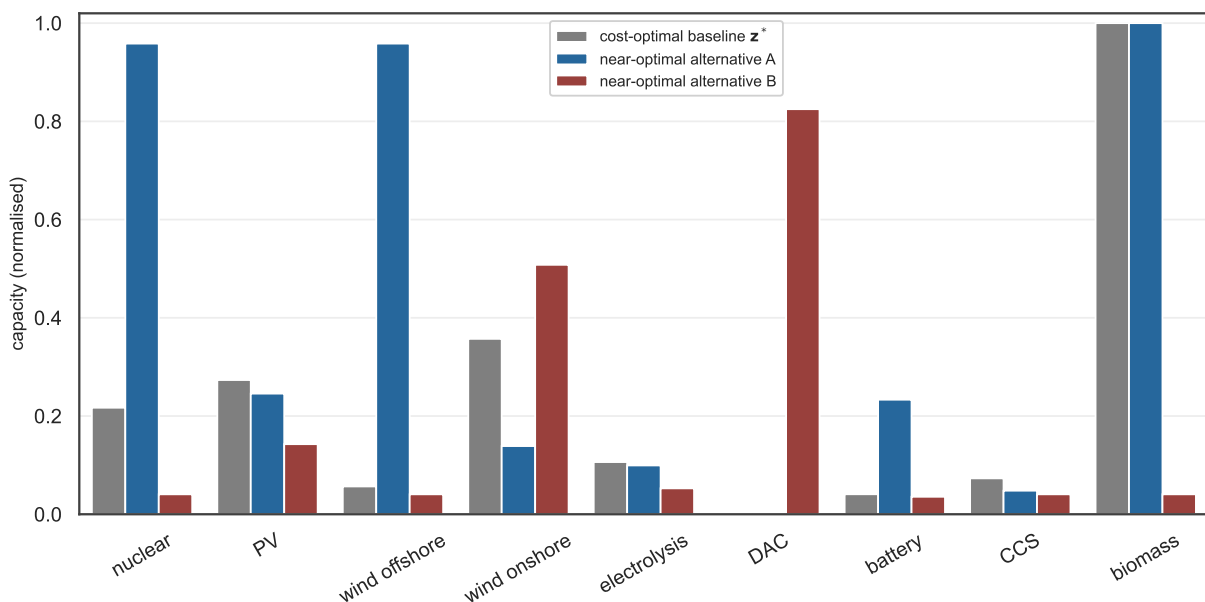


Figure 4: The cost-optimal baseline z^* and the two most dissimilar certified near-optimal designs (the pair with the largest Euclidean distance over the nine normalised design axes, among all 701 certified designs), each axis normalised by its near-optimal maximum $\bar{z}_i$ (cf. the normalised design display in [14]). All three designs cost within $\varepsilon = 10\%$ of C^* ; both alternatives lie at the top of the allowed cost band. These are certified boundary designs showing how far two near-optimal systems can diverge, not typical interior designs.

The exploration of the near-optimal space around this baseline, whose course Section 4.2 reports, **certified 701 near-optimal designs**. Figure 4 shows the baseline beside the **two most dissimilar** of them, the pair of certified designs farthest apart in the normalised design space.

The three systems supply the **same demand at almost the same cost**, yet they have little in common. Alternative **A** pushes **nuclear and offshore wind** to almost their near-optimal maxima and, like the baseline, **exhausts** the **biomass** import. Alternative **B** instead centres on **onshore wind and DAC**, and nearly renounces what the other two rely on: it imports 4% of the available biomass and builds 4% of the near-optimal nuclear and offshore-wind maxima. Both alternatives lie at the **upper edge of the allowed cost band**, so their dissimilarity is bought with the **full cost slack**. This contrast is what near-optimal exploration is meant to expose: a cost-optimal solve returns exactly one of these systems and hides the others, although they cost at most 10% more. The two alternatives are **certified boundary designs** that show **how far apart near-optimal systems can lie**; where the **bulk** of the near-optimal designs sits is the subject of **Section 4.3**.

4.2 ORACLE convergence and the near-optimal polytope

The **final exploration, run 15**, used its **full budget of 700 iterations** and roughly 61 hours of loop time on a dedicated compute server (Appendix, Table A2). **Each iteration** solved the **max-min problem** of Eq. (7), which first hit its **120-second time limit** at iteration 59 and did so in every iteration from 74 onward, and one **projection of the full model**, Eq. (8), which took about **200 seconds on average**. **Every trial point proved infeasible**, so every iteration added **one certified design to the inner approximation** and one **cutting plane** to the **outer**: the final pair holds **701 certified designs** (the cost-optimal design plus one per iteration) and **720 half-spaces** (the 20 faces of the **initial box** plus one cut per iteration) in **ten dimensions**. Two earlier explorations are reported alongside the final run: **run 8**, the early run whose **uncorrected metric readout** produced the **false convergence** that was fixed in Section 3.2, and **run 13**, which differs from the final run in its **tolerance** and its **certificate formulation**. The settings in which the three explorations differ are listed in the Appendix (Table A4).

Figure 5 traces **two quantities per exploration**: the **reported metric**, i.e. the **proven bound** of the **certificate solve**, a **valid upper bound on $d_{\mathcal{IO}}$** , and the **incumbent**, the **largest**

distance the solve actually found, a valid lower bound. The three explorations first descend along a nearly identical path: within 73 iterations, the proven bound falls from its initial value of 1 to about 0.26. From that iteration on, the proven bounds of runs 13 and 15 never improve again and plateau at 0.260 and 0.258, respectively, while the incumbents keep falling, in the final run to below 0.09. The growing gap between the two traces means that the per-iteration max-min solves stop proving what the exploration achieves: the bracket around the true distance widens, not because the approximations stop improving, but because the certificate can no longer follow them within its per-iteration budget. The final run already used the more effective of the two certificate formulations for its problem (Section 3.2), yet its proven bound plateaued in the same way. Its 120-second limit was accordingly not meant to close the bound within the loop; it was set to reliably return a valid trial point and witness even on the refined late-iteration geometry, with a safety margin, while the proof itself is bought once at the end.

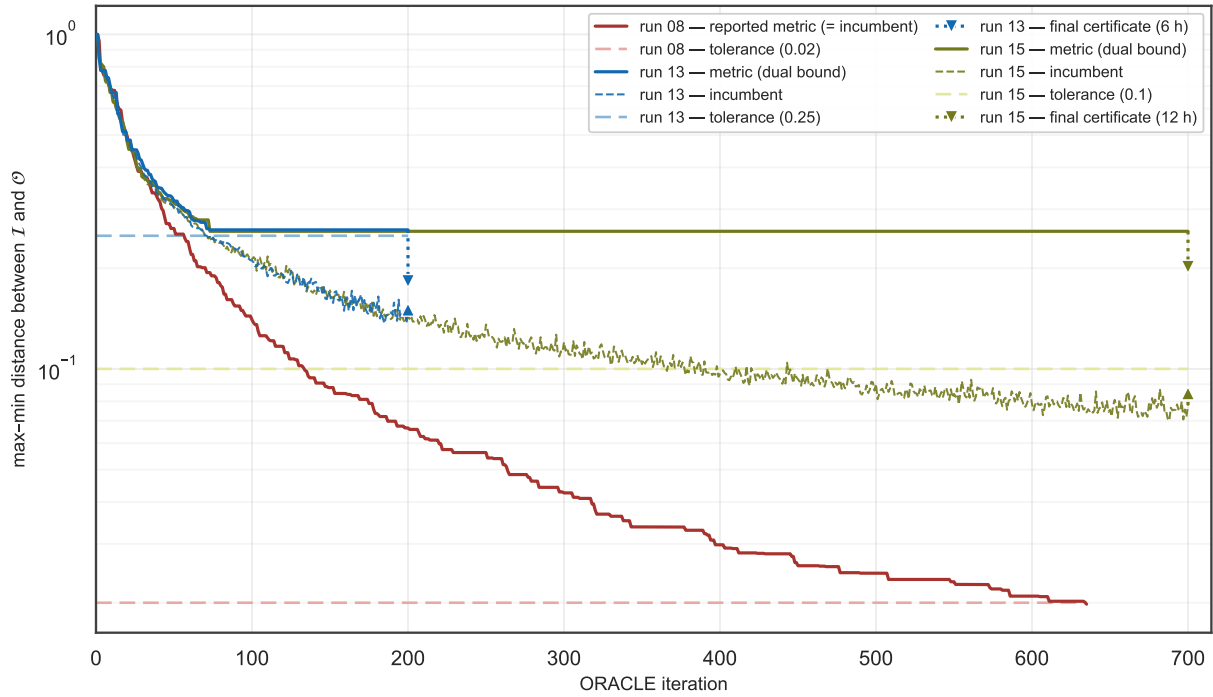


Figure 5: Max-min distance between the inner (I) and outer (O) approximations over the ORACLE iterations, runs 8, 13 and 15. Solid: the reported metric (the solver’s proven bound in the fixed runs 13/15); dashed: the incumbent (the largest distance actually found); faint dashed horizontal: each run’s convergence tolerance, drawn across its own iterations. Run 8 shows a single trajectory: its “metric” was the incumbent, the false-convergence pathology. The dotted markers at the last iteration of runs 13 and 15 show the final certificate tightening the bracket: the proven bound falls and the incumbent rises.

For the final iteration of the final run, we therefore raised the time budget of the max-min solve to 12 hours. This lowered the proven bound from 0.258 to 0.202 and raised the incumbent to 0.085: the true distance of the final approximations lies between 0.085

and 0.202. The loop’s **stopping criterion**, a proven bound **below the tolerance of 0.1**, was thus **never met**, and the exploration **did not terminate as converged**. Whether the final approximations nevertheless satisfy the tolerance remains **undecided**: **certified** is only that the **distance does not exceed 0.202**, while its **true value may well lie below 0.1**. The **two earlier explorations isolate** the **two requirements** a **trustworthy metric** must meet. It must first be **valid**. **Run 8** predates the corrected readout and reported the incumbent as its metric; its single trajectory in Figure 5 descends smoothly to 0.0198, just below its 0.02 tolerance, and the run stopped as converged. A **later certificate** on the same geometry closed its gap exactly and placed the **true distance at 0.276**, roughly fourteen times larger. The **reported convergence was false**, because an **incumbent bounds** the distance **only from below**; the metric must be read as the **proven bound**, as runs 13 and 15 do. A valid bound is not yet a tight one. Both later runs stalled in the loop at their time-capped budgets alike, so we **recomputed the final certificate afterwards on run 13’s finished geometry**, once with each reformulation and each given six hours. The published **KKT-MILP** explored **4.4×10^7 branch-and-bound nodes without improving** on 0.260, whereas the **strong-duality reformulation** lowered the proven bound to **0.183** against an incumbent of 0.150 (Appendix, Figure A2). The strong-duality bound certifies run 13 as converged at its 0.25 tolerance and **motivated using this reformulation for the final run**.

4.3 Interpreting the near-optimal space

All **sample-based results** in this section rest on the **rejection sample of run 15**. Of the **9.3 million uniform proposals** drawn from the **outer body**, **11.6%** lay in the **inner body**; this **acceptance rate** estimates the **inner body’s share of the outer volume** (Eq. (13)). The **eight walks are statistically indistinguishable** ($\text{split-}\hat{R} \leq 1.0001$ on every axis), the retained **sample** is worth at least **150 000 independent draws per axis** (minimum bulk ESS, target 100 000), and the **independent direct walk on the inner body agrees** with the **rejection sample** on every axis and every axis pair. All statistics below use the **full retained set of 1.06 million uniform draws** from the **inner body**, each of which is a certified near-optimal design.

Necessary conditions

The certified minima and maxima of the nine design axes, the three aggregates and the two conditional bounds are given in Table 1 (Section 3.4). Every single axis has a certified minimum of zero (at solver precision): no explored technology is individually indispensable, each can be omitted entirely at a cost premium of at most 10%. In the terminology of [20, 35, 36], all nine axes are *Real-Choices*. The aggregates, in contrast, are *Must-Haves*: every near-optimal design contains at least 1480 GW of VRE capacity and at least 15.6 ktCO₂eq/h of CO₂-capture capacity, of which at least 8.3 ktCO₂eq/h must be CO₂-removal capacity. Every near-optimal design must therefore provide bulk renewable supply, capture capacity and a core of negative-emission capacity, even though no single technology delivering any of them is indispensable. Computing these bounds on the full model is what makes them exact: read off the final approximations instead, the capture minimum would be 0 (outer body) or 31.5 (inner body) rather than 15.6, all in ktCO₂eq/h, because no cut happened to certify this floor on the outer body, while the inner body under-covers the space.

Table 1: Certified necessary conditions of the near-optimal space: minima and maxima of the nine design axes, three aggregates and two conditional bounds, each an exact solve of the full model subject to $\mathbf{c}^\top \mathbf{x} < (1 + \varepsilon)C^*$, computed directly rather than read off the polytope approximations. A strictly positive certified minimum makes a row *Must-Have*; a minimum of zero (at solver precision) leaves it a *Real-Choice*. Per-axis maxima are the near-optimal maxima $\bar{z}_i$ of Eq. (11); the conditional rows pin one axis to zero and bound the other. The aggregate members are defined in Section 3.4. Values are given to one decimal place.

Axis	Unit	Certified min	Certified max	Class
Nuclear	GW	0.0	110.4	Real-Choice
PV	GW	0.0	9834.9	Real-Choice
Wind offshore	GW	0.0	1149.0	Real-Choice
Wind onshore	GW	0.0	4351.1	Real-Choice
Electrolysis	GW	0.0	2688.5	Real-Choice
DAC	ktCO ₂ eq/h	0.0	91.4	Real-Choice
Battery	GWh	0.0	18 286.2	Real-Choice
CCS	ktCO ₂ eq/h	0.0	843.5	Real-Choice
Biomass import	GWh	0.0	777 923.7	Real-Choice
Total VRE	GW	1483.3	10 951.6	Must-Have
CO ₂ capture	ktCO ₂ eq/h	15.6	843.6	Must-Have
CO ₂ removal	ktCO ₂ eq/h	8.3	810.6	Must-Have
Biomass DAC=0	GWh	542 241.4	777 923.7	Must-Have
DAC Biomass=0	ktCO ₂ eq/h	24.5	80.7	Must-Have

The two **conditional bounds** extend the requirement to a pair of axes. With the **biomass** import pinned to **zero**, no near-optimal design builds less than 24.5 ktCO₂eq/h of DAC; with **DAC** pinned to **zero**, none imports less than 542 TWh of biomass, **70% of the available resource**. Each of the two axes may be abandoned on its own, but **not both at once**: every near-optimal design commits to at least one of the model’s negative-emission routes at scale.

The **certified maxima** bound the ranges upward. Even among the axes that share the **unit GW** they span **two orders of magnitude**: within the cost band, the model accepts up to 9835 GW of PV, but at most 110.4 GW of nuclear. Which **combinations of extremes** are **jointly attainable** is shown by the **pairwise projections** of the two approximations in the Appendix (Figure A4).

The distribution of near-optimal designs

Figure 6 shows where within these bounds the near-optimal **volume actually lies**. None of the marginals shows more than one peak, and **most of the volume sits far from the certified extremes**. Seven of the nine axes have **right-skewed marginals** that peak at 5% to 29% of their respective maxima; on each of them, 95% of the volume stays below 57% of the certified maximum. The **two exceptions** sit at **opposite** ends. The **biomass import is the only axis whose volume concentrates near its maximum**, the availability limit of the dataset: half of all draws import more than 76% of it, and 44% import more than 80%. **Nuclear**, in contrast, **spreads almost evenly**: the deciles of its range are nearly equally occupied (the first tenth of the range carries 10% of the draws, the last tenth 3%), so within its small certified range, **every nuclear level is almost equally represented** in the near-optimal volume.

The **pairwise panels** add the joint picture. In each panel the mass sits well inside the projected inner body (green outline) and **gathers in one region**, towards the **lower end of most axes but towards the upper end of the biomass import**, thinning out towards the opposite corners. Several pairs additionally show a **cut upper-right corner**, among them PV with onshore wind, DAC with electrolysis and DAC with the CCS lump, so **two such axes cannot approach their maxima at once**. In the biomass–DAC panel, the **dense** region combines **low DAC with high biomass** import; towards **larger DAC values the volume thins rapidly** (draws above half the DAC maximum make up 3.6% of the volume) and

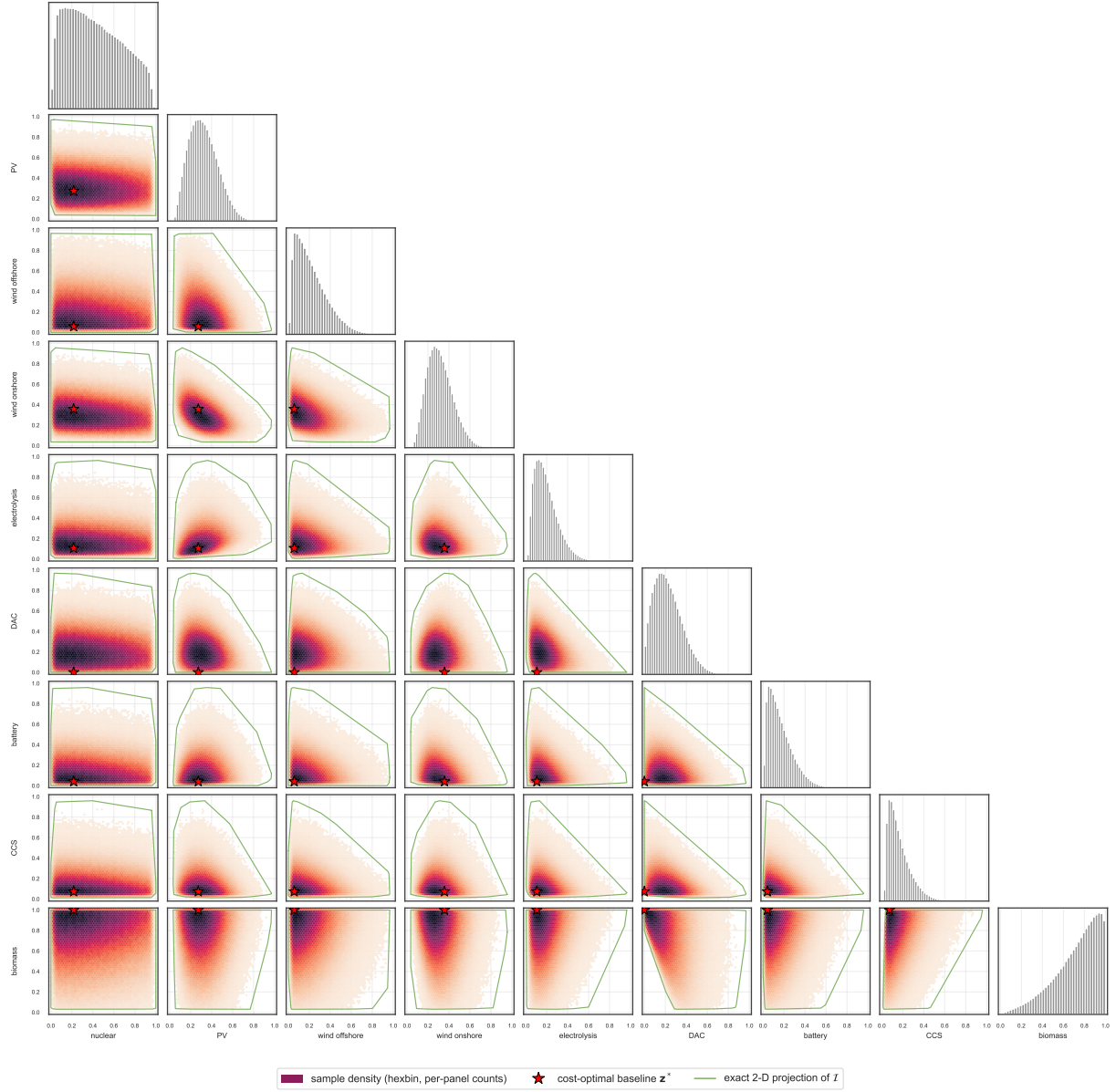


Figure 6: **Pairwise density** of the **uniform sample** on the inner approximation $\mathcal{I}$ (run 15; axes normalised by their near-optimal maxima). **Lower triangle**: **hexbin density** over all draws per axis pair; **diagonal**: **per-axis marginal distributions** of the full sample set; green outlines: the **exact 2D projections** of $\mathcal{I}$; the star marks the cost-optimal baseline $\mathbf{z}^*$. **Darker cells carry more of the near-optimal volume**.

shifts to lower biomass imports (median 0.63 of the biomass maximum, against 0.84 where DAC stays below one fifth). In the **nuclear** column, the **density bands run horizontally**: the **distribution of every other axis looks alike at every nuclear level**. A scatter view of the same sample set, with all certified designs and the baseline overlaid, is provided in the Appendix (Figure A3).

Pearson correlation

Figure 7 quantifies the **pairwise couplings**. **All coefficients are moderate at most**: none exceeds 0.45 in magnitude, and most are **negative**, meaning that where **one technology is built out more**, others tend to be built out less.

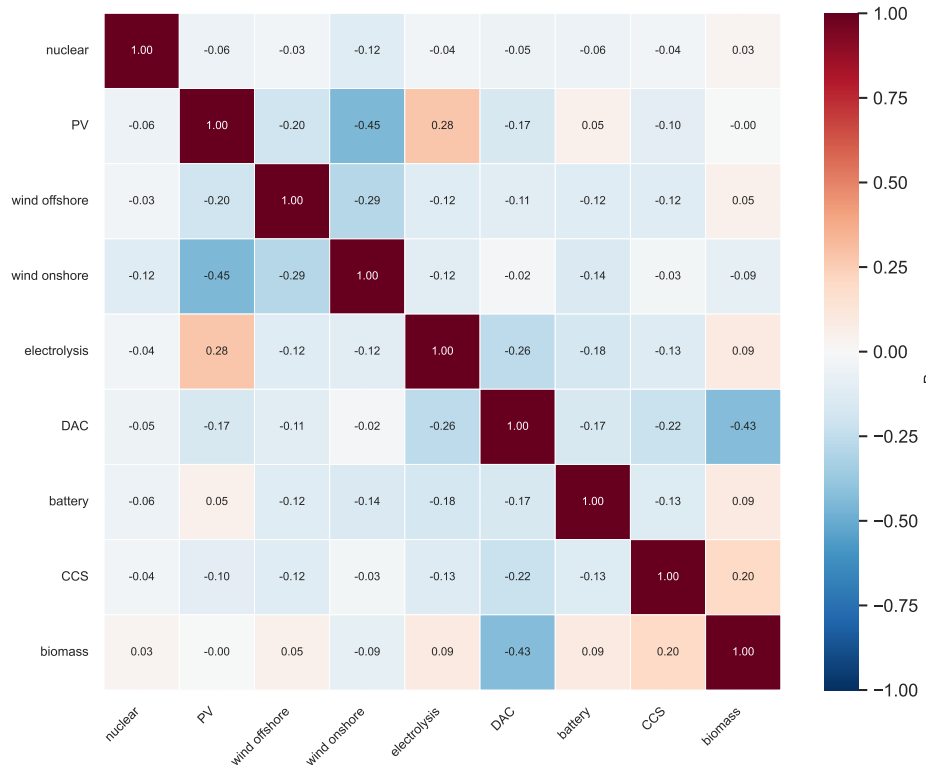


Figure 7: Pearson correlations between the design axes under the uniform measure on the inner approximation $\mathcal{I}$ (run 15, full sample set). The **dependence is essentially linear** (Appendix A2), so the Pearson matrix carries the pairwise story; pervasive **negative correlations reflect substitution** within the near-optimal volume.

The **two strongest couplings** are both **substitutions**: **onshore wind against PV** (-0.45) and **DAC against the biomass import** (-0.43). The **three VRE axes substitute pairwise throughout**, onshore against PV as above, onshore against offshore wind at -0.29 and PV against offshore wind at -0.20 , so the renewable supply can shift between them. **Electrolysis rises with PV** ($+0.28$), the largest positive coupling, and **declines** mildly with

DAC (-0.26) and battery storage (-0.18). The CCS lump stays weakly coupled to every axis, reaching at most $+0.20$ with the biomass import and -0.22 with DAC. Nuclear shows no appreciable coupling with any axis ($|r| < 0.13$), matching its flat marginal and horizontal density bands in Figure 6. The biomass import feeds several conversion routes, so what the DAC–biomass substitution means for the underlying system is taken up in the Discussion.

These linear coefficients carry the pairwise structure of the sample. An OLS regression of every axis on the remaining eight caps the multivariate picture: no axis is more than 56% linearly explained by the rest (Appendix, Figure A6). Spearman rank correlations match the Pearson values closely, and neither mutual information nor distance correlation reveals any pair with appreciable dependence that the linear screen misses (Appendix, Figure A5). A principal-component analysis (PCA) of the draws spreads the variance over several directions, the leading one carrying 30% (Appendix, Figure A7).

5 Discussion

The previous chapter established **two principal results**: the exploration built a **polytope pair** whose **true distance** is certified to lie between **0.085** and **0.202**, and the bracketed near-optimal space combines **necessary conditions** on its **aggregates** with wide **freedom in the individual technologies**. This chapter interprets both. Section 5.1 discusses what the observed **convergence behaviour means for ORACLE as a method**. Section 5.2 interprets the **structure of the near-optimal space** and the **mechanisms** behind it. Section 5.3 assesses the **cost coordinate**, and Section 5.4 states the **limitations** of this work.

5.1 ORACLE as a general method

The exploration had **two parts** that **behaved** very **differently** as the **problem grew**. The **refinement** itself **held up**: the loop completed **all 700 iterations**, the incumbents kept falling, and **every cut removed further volume from the outer approximation**. The **certification** did **not** keep pace: the **dual bound** of the per-iteration solves **stopped improving** within their budget after 73 iterations, and even the twelve-hour final certificate left the bracket astride the tolerance. The cause is **not** the **size of the underlying model**, as the following makes precise, but above all the **number of explored dimensions**.

The certification bottleneck. The reason the **certificate** hardens over a run is that it **grows with the exploration**. The **max-min problem** of Eq. (7) is **solved over the two approximations alone**, so its **size is set by the number of dimensions, certified points and cuts**, not by the model behind them. Each iteration adds one certified point and, whenever the **trial point is infeasible**, one cut, so the problem the certificate must solve is **largest at the end of a long run**. The published MILP reformulation makes the dependence explicit, with $2n_z + m$ binary variables for m certified points [14]. The **two final certificates** show the effect: run 13, with 201 points, was lowered from 0.260 to 0.183 in six hours, within 0.033 of its incumbent, whereas run 15, with 701 points, was lowered from 0.258 only to 0.202 in twice the time, still 0.117 above its incumbent. The **model's 467 280 variables enter only through the projection** of Eq. (8), **one solve per iteration**, and appear nowhere in the certificate. A **larger or more detailed model** would **raise the cost of each projection** and could, through a more intricate near-optimal boundary, call for more points; but the **point count is driven first of all by the explored dimension**, as the comparison below

argues.

This difficulty is expected for problems of this type. Bilevel linear programs are strongly NP-hard, and no fully polynomial approximation scheme exists for them unless $P = NP$ [25]. The published reformulation through the KKT conditions turns each constraint of the inner problem, and hence each certified point, into a complementarity condition that the solver branches on [28], and such formulations are known to yield weak bounds [37]: on run 13’s final geometry, the KKT-MILP explored 4.4×10^7 nodes without improving its bound at all. Branch-and-bound moreover spends most of its effort on the proof rather than on finding the optimum; even when the optimum is supplied in advance, the greater part of the search remains [25]. The final run shows this directly: its incumbent of 0.085 would satisfy the tolerance, but the proof stops at 0.202. This motivated the strong-duality reformulation used for the final run, which replaces the binary variables by a single bilinear constraint [28] and, on our geometries, reached tighter bounds than the KKT-MILP in equal time. It is not a general remedy, as it remains a non-convex global optimisation problem and did not close run 15’s gap either.

Why this reformulation closes some geometries and stalls on others, we can offer only a tentative reading. It closed the corrupted run-8 geometry almost immediately, returning the exact distance 0.276 with no remaining gap. A plausible explanation is that a single region of the outer approximation, left uncut by the defective run, lay far beyond all others; ruling out the rest was then easy. On the healthy geometries no such dominant region exists. After several hundred cuts, many parts of the outer approximation are almost equally far from the inner one, and the solver must exclude each of them before the dual bound can fall past this cluster of near-equal distances. The difficulty of closing such a bilevel bound is indeed known to vary by orders of magnitude between otherwise comparable instances [25], and this reading is consistent with the shape of the certificate traces, which make fast initial progress and then improve only slowly (Appendix, Figure A2).

Scaling with dimension. The comparison with the original ORACLE study points to the explored dimension as the main driver. The published case study explored six axes of a spatially aggregated, single-node model of Switzerland with fast solves [14], and certified an absolute tolerance of 0.1 GW, about 0.25% of the initial max–min distance of roughly 40 GW, within 144 iterations. We explored ten axes of a 28-node sector-coupled model and

targeted a tolerance of 10% of the initial distance, forty times coarser in relative terms, yet needed 700 iterations to bring the incumbent near it and could not certify it. The larger model raises the cost of each projection, but the gap in iteration count and provability points elsewhere: approximating a convex body to a fixed relative tolerance can, in the worst case, require a number of vertices and facets that grows steeply with the dimension [38, 39, 40], so a ten-dimensional bracket plausibly needs far more certified points and cuts than a six-dimensional one, and the certificate inherits this growth. The projection of Eq. (8), by contrast, cost about the same in every iteration, because it depends on the model and not on the growing approximations; that this constant stayed near 200 seconds owes much to the coarse temporal resolution of 10 representative time steps. Its absolute solve time is therefore immaterial to the trend: over a run, only the certificate's cost grows, in line with the original study, which identifies the number of iterations and dimensions, rather than the model, as what governs the cost of the max-min step [14]. For the number of explored dimensions, this work sits at the upper end of what has been reported. The studies that construct the near-optimal space as a region (Section 2.2) have not exceeded ten dimensions: the explicit hulls of MAA become infeasible above ten even on large hardware [12, 13], and the published applications use five to seven [13, 14, 15]. MGA runs with more dimensions exist, for instance in the capacity-expansion experiments of [12], but they generate boundary designs and estimate the explored volume statistically rather than constructing the region and bounding it from both sides. Dimension pressure affects the two region-constructing methods differently. In MAA it breaks the search itself: without the explicit hull there are no facet directions to explore and no volume to monitor. In ORACLE the search survives, because a time-capped max-min solve still returns a valid trial point, from which projection and cut proceed as usual [14]; only the tightness of the certificate degrades. Our runs are an instance of exactly this behaviour, with refinement continuing for 700 iterations while the certificate stalled.

Normalisation bears on the certificate's difficulty as well. With axes in four different units, normalising is the only way to make one tolerance meaningful for all of them, and the near-optimal maxima are the natural scale (Section 3.2). It also changes how the axes enter the certificate. Under a single absolute tolerance, as in the original study, an axis whose whole range is smaller than the tolerance cannot affect the max-min distance at all: unnormalised, the nuclear axis with its range of 110 GW would be negligible beside the

biomass axis with its 777 924 GWh, and would effectively **drop out of the certification**. The **certification would then converge without ever resolving that axis**, so the unnormalised alternative is easier because it **certifies less**. Normalisation removes this effect, so that **all ten axes carry equal weight and each must be resolved** to 10% of its own range. This makes the **certification genuinely ten-dimensional**, which plausibly leaves more near-equal distances for the solver to discriminate. Whether this equal weighting adds difficulty beyond the raw dimension count, we cannot separate from our runs; it could be **tested by exploring ten axes that all share one unit and similar magnitudes**, so that normalisation makes no difference, and comparing the certificate against the normalised run.

The guarantee and the bracket. None of this contradicts the method’s convergence guarantee. The **paper guarantees convergence to within a chosen tolerance** [14], on the premise, left implicit, that **every max–min solve is closed to proven optimality** (Section 3.2). It is exactly this premise that **fails once the dimension grows**: the guarantee does **not break**, but it **depends on a strongly NP-hard problem being certified at every iteration**. The **stopping rule fails with it**: because the loop can no longer prove when it may stop, the **pre-set iteration budget**, the very device that certified convergence was meant to replace (Section 2.1), **returns as the operative stopping rule**. What remains, and what we consider the method’s real asset in this regime, is that a stalled certificate still yields a **valid two-sided bracket on the coverage error**. Reported as the **pair of incumbent and dual bound**, this bracket is a **coverage statement that no other MGA method provides**. The practical recipe that follows is to **refine with cheap, time-capped max–min solves and to buy the proof once, on the final geometry**. Options for tightening the bracket further are discussed in the Outlook (Section 6).

5.2 Insights into the greenfield-2050 near-optimal space

The results separate two levels of statement. Every axis can be driven to zero within the cost band, yet no near-optimal design avoids 1480 GW of VRE or 15.6 ktCO₂eq/h of total capture capacity. The **near-optimal space therefore constrains the system’s aggregate composition**, its bulk renewable supply and its total capture, while **leaving open which technologies provide them at a cost premium** of at most 10%. This has two **consequences for planning**. Commitments made at the level of aggregates hold across the whole near-optimal space. And removing any single explored technology from the build-out does not

by itself push the system out of the near-optimal band, because the remaining technologies can take over its role.

A flexible interior. Interpreting the **interior statistics** requires care about what the **uniform measure** means. A region of **high sample density** holds a **large share of the near-optimal volume**, meaning that **many distinct capacity combinations realise it**. Pedersen et al. read such density as **implementability**, arguing that **dense regions “are more likely to be implemented as they can be achieved under several configurations** of the energy network” [13, p. 9], whereas Lau et al. warn that a design’s position within the near-optimal space bears **no relation to its real-world feasibility or likelihood** [11]. We adopt the cautious reading (Section 2.2): density measures **how much room a region leaves for reconfiguration within the model, not how probable its realisation is**. In this restricted sense, an **axis is flexible when fixing it still leaves a wide range of near-optimal values open on the other axes**. **Weakly coupled** axes have exactly this property: fixing one of them barely constrains the rest.

The near-optimal space is flexible in exactly this sense. **No pairwise correlation exceeds 0.45** in magnitude, no axis is more than 56% linearly explained by the others, and the **sample variance spreads across several principal directions rather than collapsing onto one**. Within the 10% cost band, the **technology choices are therefore largely independent** of one another, in line with the flexibility that interior studies of near-optimal power systems report [13, 15, 20]. This flexibility is a property of the space itself, not of the particular exploration method, on two counts. Because **every interior sample is a certified near-optimal design, the observed freedom is a lower bound on the truth**. And because the bracket bounds what the exploration may have missed, the freedom is not unboundedly larger either: **no axis reaches beyond the explored body by more than the certified distance**. An exploration method that returns only a handful of boundary designs can bound neither side, and Lau et al. show how far such a method can understate the freedom that is actually present [12].

The **interior** also **corrects the ranges**. On seven of the nine axes, 95% of the near-optimal volume lies below 57% of the certified maximum: **what is possible at the boundary is not what the bulk of the space looks like, and ranges alone over-read the space** [11]. The biomass import is the instructive exception. It is the only axis whose volume concentrates at its upper bound, with half of all designs importing more than 76% of the available

resource: at the dataset’s 2050 import price of 34.1 EUR/MWh, biomass is attractive enough that most near-optimal systems exhaust it. The availability limit, an exogenous dataset assumption, thereby shapes the entire characterisation (Section 5.4).

Capture and negative emissions. We read the coupling between DAC and the biomass import (-0.43) as competition between the model’s two routes to negative emissions. The reading follows from how the model accounts for carbon. Biomass is imported at zero accounted emissions [24], because the carbon in it was taken from the atmosphere as the biomass grew; releasing it again through combustion is likewise uncounted; and carbon that is captured and sent to storage is booked as a removal. The full path from the atmosphere through biomass, its combustion, and capture to storage therefore leaves the system with a net removal, and DAC achieves the same by capturing atmospheric carbon directly. Fossil fuels, by contrast, are debited with their carbon content when imported, so capturing and storing their emissions only cancels that debit and nets to roughly zero. Biomass on its own is thus carbon-neutral, point-source capture on fossil fuels prevents emissions rather than removing them, and genuine negative emissions arise only along the two removal routes: biomass whose carbon is captured and stored, and DAC. Under the fixed emissions constraint, the removals can come from either route, so designs that build more DAC import less biomass. The biomass import also lowers how much removal is needed in the first place: used instead of fossil fuels, it avoids residual emissions that the removal routes would otherwise have to offset. Either way, more biomass means less DAC.

The necessary conditions certify this competition as a requirement rather than a tendency of the volume. The removal aggregate cannot fall below 8.3 ktCO₂eq/h in any near-optimal design, and the conditional bounds of Table 1 show the two routes standing in for each other: closing the biomass route forces at least 24.5 ktCO₂eq/h of DAC, about a quarter of its certified maximum, while forgoing DAC forces at least 70% of the available biomass import. This joint floor is also visible as geometry: the cut lower-left corner of the biomass–DAC panel in Figure 6 is exactly the region these bounds exclude.

The correlations bear this out as far as the axes allow. The biomass route’s own capture capacity sits inside the CCS lump, whose five other members capture fossil and process emissions, so the lump is dominated by prevention rather than removal. Its only two non-negligible couplings fit the mechanism: a positive $+0.20$ with the biomass import, its sole

positive coefficient, as the biomass-fed member grows with the resource, and a negative -0.22 with DAC, a weakened form of the removal substitution that reaches the lump through that one member. Both are weak, but consistent with the reading rather than with noise. A sharper attribution is possible in principle: the plugin stores the complete design behind every certified boundary point, so the biomass allocation of high-DAC and low-DAC designs could be compared to separate the members of the lump. We have not done this, and for the interior samples it is not possible, since a sample records only the axis totals, which cannot be split among the lumped technologies [7].

The capture Must-Have, the certified minimum of 15.6 ktCO₂eq/h of total capture capacity (Table 1), plausibly originates in unavoidable industrial process emissions. The model’s clinker and steel demands are exogenous, and the CO₂ released when limestone is calcined into cement clinker is chemically unavoidable: it cannot be removed by switching energy carriers [41], only captured at the source or offset by removal, both of which need capture capacity. This is why the floor appears on the aggregate while every individual capture technology stays omissible. The exogenous clinker demand alone implies of order 7 ktCO₂eq/h of capturable kiln emissions, about half the floor; the remainder is not fixed by this argument, since steelmaking can avoid the bulk of its process emissions by switching to hydrogen-based reduction [42]. A full attribution would require the emission balance of the capture-minimising design, which we did not record.

Caps and couplings. Nuclear shows how strongly a single interpretation can hinge on the dataset. Its certified maximum of 110.4 GW is not an economic outcome: the dataset limits total nuclear capacity to 119.1 GW across the fifteen countries permitted to build it, and the certified maximum together with the 8.7 GW surviving from the 2022 fleet reaches this limit exactly. The model was therefore never free to expand nuclear on a large scale, and within its permitted range nuclear is too small to influence a system whose PV axis alone spans 9835 GW, which accounts for its flat marginal and its near-zero couplings. A statement such as “nuclear plays no role in the 2050 system” would merely restate this exogenous limit. Other axes are capped as well, but the caps differ in character and not all of them bind. The limits on offshore wind and on biomass imports, which the near-optimal maxima also reach, are most naturally read as resource ceilings, set by the available coastal sites and the sustainable biomass supply, so that reaching them is itself informative. The nuclear limit is low relative to any role the technology might plausibly

play and is better read as a modelling assumption. PV and onshore wind stay a few percent below their potentials, so their maxima are set by the system rather than by a cap, and electrolysis, DAC and battery storage have no finite cap and are bounded by cost alone.

The cost band bounds pairs of axes as well as single ones. Many of the pairwise panels show a cut upper-right corner, where two axes cannot approach their maxima together (Section 4.3). Reaching a certified maximum spends much of the cost slack on that one axis, so a second axis cannot also be driven to its own maximum without leaving the cost band. These corners therefore record the cost budget, not a physical ceiling on the pair. The remaining couplings read naturally. The three VRE axes substitute pairwise (-0.45 to -0.20 , weakest between PV and offshore wind): the aggregate floor demands bulk variable renewable supply, and which source provides it is secondary, matching the observation that near-optimal renewable investment can be shifted between technologies [15]. The largest positive coupling, PV with electrolysis ($+0.28$), matches the kind of positive coupling reported between generation and the technologies that absorb its output [7]; solar-powered hydrogen production is the plausible mechanism, though we have not traced it in the model's flows.

In sum, the near-optimal space of the greenfield-2050 system pins aggregates and frees technologies, its volume concentrates well inside the certified ranges, and its moderate couplings follow identifiable mechanisms: resource competition for negative emissions, process emissions demanding capture capacity, and substitution within the renewable supply.

5.3 Cost as an exploratory variable

The cost coordinate served one purpose in this work: making cost part of the explored geometry. No cost-based analysis was conducted, and all reported statistics live on the nine-dimensional design projection (Sections 3.2 and 3.3). The explored design space itself does not change: dropping the cost coordinate from the ten-dimensional body recovers exactly the nine design axes, since both are images of the same $\mathcal{X}_\epsilon$, so including cost neither adds nor removes any near-optimal design.

What the coordinate changes is the visibility and control of cost. Every certified design now carries its own cost value, so the way the cost slack is spent, previously left to

the solver, becomes an explored quantity, and each projection solve is steered towards a prescribed cost level rather than an arbitrary one. Standard weighted MGA cannot do this. Its cost constraint only bounds cost from above, and within that band the objective is indifferent to cost, so the cost of a returned design is whatever the weight vector happens to produce, and the slack may be spent invisibly on quantities the axes do not track; in a pathway test run it was absorbed by the carbon-overshoot penalty (Section 3.2). Adding a weight on cost does not resolve this, it merely pulls every solve to the cheap edge of the band and leaves the expensive part unmapped. Carrying cost as an explored coordinate instead maps the whole band, and ORACLE’s implicit hull absorbs the extra dimension where an explicit-hull method would pay the Quickhull price for it (Section 2.1).

The extra dimension is computationally cheap, but it is not free for the analysis: it biases the projected design marginals, which the nine-dimensional sampling corrects, and it spends refinement effort on a tenth dimension (Section 5.4). Against this stand analyses that only the cost dimension enables and that this work leaves unused. With cost in the polytope, the cheapest near-optimal design meeting given capacity preferences, the minimum extra cost of pushing one technology, or cost-conditional design statistics all become LPs on the converged body, with no further model solves. Whether such analyses are best served by the 10% band of this work or by a wider one is open: empirical hindcasts report realised cost deviations well above 10% [43], but how faithfully a model’s near-optimal band represents such deviations is itself debated.

5.4 Limitations

The first group of limitations concerns the model configuration. Crystal Ball was built to model the European transition as a pathway, and this work re-purposes it for a one-shot 2050 design by fixing a single target year and disabling the technology-diffusion limit, the year-on-year growth constraint that is specific to a pathway (Section 3.1). Studying a single target year is common in the near-optimal literature, and disabling the diffusion limit removes the one pathway-specific constraint that would otherwise forbid it; the remaining input assumptions, the technology potentials and resource limits, are static and apply to a greenfield design as well. Some of them nonetheless shape the results and deserve scrutiny in this context: the biomass availability limit, at which most of the near-optimal volume sits, and the nuclear capacity limit, which the certified maximum reaches

exactly, both discussed in Section 5.2. The further caveats are the inherited long-lived infrastructure, the **coarse temporal resolution of ten representative time steps per year**, and **exogenous 2050 technology costs without endogenous learning**.

The second group concerns the **exploration** itself. The nine design axes cover a chosen **subset of the model’s technologies**, selected as the most debated design choices, while all **other decisions** remain **unexplored background**. Each axis sums one technology, or several in the case of the CCS lump, over all nodes, so the axis total is exact at the technology level but does **not resolve where capacity is sited**, and for the lump it does not separate the constituent technologies [7]. The **slack $\varepsilon = 0.1$ is a subjective setting**, and **empirical estimates of realised cost deviations are far larger** [43]; every reported range and volume scales with it. The exploration also spends **refinement effort on the cost axis**, so the projected nine-dimensional approximations are less refined than those of a dedicated nine-axis run at equal iterations.

The third group concerns **coverage**. The **certification is a bracket, not a point**: the **true max–min distance of the final approximations lies between 0.085 and 0.202**, straddling the 0.1 target, so **convergence** at the target tolerance was **neither certified nor refuted** (Section 4.2), and every certified range inherits this bound. The same gap appears in volume terms: all **volume statistics are read off the inner body, which under-covers the true region**, and the **inner body fills 11.6% of the outer one**. If that volume gap were spread evenly over the nine axes, it would correspond to a **factor of $0.116^{1/9} \approx 0.79$ per axis**, consistent with the certified **per-axis bracket**; the **true region lies between the two bodies**. **Sampling** would also **grow harder with dimension**. The rejection sampler’s acceptance rate is exactly this volume ratio, so at comparable bracket quality it falls geometrically as axes are added, and ever more proposals are needed for each accepted draw. The **direct walk on the inner body** does **not depend on that acceptance rate**, so beyond some dimension it may overtake rejection as the faster of the two, but it **carries its own cost**, **pricing two LPs at every step** and mixing **slowly**, which already made it several times slower here (Section 3.3). Either route grows more expensive with dimension; only which is slower changes.

6 Conclusion and Outlook

Before this work, ORACLE had been demonstrated only on a six-dimensional, single-node model. We applied it to a ten-dimensional, sector-coupled model of the European energy system, asking whether its certified exploration carries over to this larger and higher-dimensional setting, and what the explored space reveals about the 2050 design. The exploration transferred cleanly: every iteration added a certified design to the inner approximation and a cut to the outer, steadily tightening the two around the near-optimal space. Proving how tightly they enclosed it was the harder part. Granted twelve hours, the final certificate placed the coverage error between 0.085 and 0.202, so the target tolerance of 0.1 was not provably met. The certificate nonetheless yields a valid upper bound: the error of reading any result off the approximations is certified not to exceed 0.202. This difficulty is intrinsic to the bilevel certificate and grows with the number of explored dimensions. At ten dimensions, this work also sits near the upper limit of what region-based near-optimal methods have so far reached [12, 13].

For the European 2050 system, the near-optimal space draws a clear line between requirement and choice. Two requirements stand out, both holding across every design within the 10% cost band: at least 1480 GW of variable renewable capacity and at least 15.6 ktCO₂eq/h of CO₂-capture capacity. Neither pins down a technology, since any single technology or import can be dropped, but the totals themselves are unavoidable. Beyond these floors the space is only loosely constrained: most of the near-optimal volume sits well inside the certified ranges, and the explored technologies stay at most moderately coupled, so fixing one still leaves the others a wide range. A single least-cost design conceals this structure and reports only one admissible system among many. The distinction is what makes the analysis useful to a decision-maker: the aggregate floors are firm, model-based requirements, while the technology mix that meets them can be chosen on grounds the model does not represent, such as public acceptance, political feasibility and competition for land [4, 8], without leaving the near-optimal space.

A large, sector-coupled near-optimal space can be characterised rigorously and from both sides; the price of that rigour at ten dimensions is a convergence certificate that no longer closes in practical time, leaving a valid coverage bracket in place of a proof. Within that bracket, the 2050 European system is firmly constrained in its aggregates and strikingly free in its individual technologies.

Looking ahead, the most pressing task is to close the certification gap. Tighter reformulations of the bilevel bound or solvers built for it would target it directly, and dropping the cost coordinate, where no cost-conditional analysis is planned, spares the certificate an additional dimension. Certification can also be separated from the loop, in both timing and cost. Each in-loop max–min solve only has to return a valid trial point, so a short time cap keeps refinement cheap, and the proof is computed separately. Because the inner approximation only grows and the outer one only shrinks, the true distance between them can only fall as the exploration proceeds, so a bound proven on an intermediate geometry stays valid for every later iteration, and the smaller mid-run problems admit tighter proofs, as the contrast between the two final certificates suggests (Section 5.1). A further option is to preface the loop with a VMM pass, seeding the inner approximation with its extreme designs and the outer one with the bounds they certify. For little more than the min/max solves themselves, this would make the axis ranges read off the approximations exact rather than approximate, a sharper characterisation for essentially the same computation (Section 4.3). Beyond the method, the converged approximations carry analyses this work did not pursue: with cost among the coordinates, the cheapest near-optimal design meeting stated capacity preferences, or the least added cost of expanding one technology, become LPs on the final approximations. Queries this cheap could also serve interactive exploration, letting decision-makers probe the near-optimal space themselves rather than read a fixed set of figures [11]. And repeating the exploration at wider cost slacks would reveal which of the requirements found here endure as the system is allowed to move further from least cost.

Appendix

A1 Input data and run configuration

This appendix collects the concrete configuration of the model, the solver, the ORACLE runs and the interior sampler; all values are taken from the run artefacts and the dataset configuration files.

A1.1 Configuration tables

Table A1: **Configuration** of the **Crystal Ball greenfield-2050 model**.

Setting	Value
Nodes / carriers	28 / 31
Technologies	84 (68 conversion, 6 CCS retrofit, 5 storage, 5 transport)
Horizon	single reference year 2050, perfect foresight
Temporal resolution	10 representative time steps per year (TSA)
Technology diffusion limit	disabled (base dataset: pathway-calibrated, e.g. 0.13 for DAC)
Carbon policy	emissions budget 23.15 GtCO₂eq ; overshoot price 5000 EUR/t annual emissions limit 0 in 2050; overshoot price 5000 EUR/t
LP size (solver-reported)	467 280 variables, 733 892 constraints

Table A2: **Gurobi settings** for the full-model LP solves of the **final run** (baseline, per-axis maxima and projections); the necessary-condition solves reuse the same options on different hardware.

Option	Value
Gurobi version	13.0.1
Hardware	AMD EPYC 9124 (16 cores), 10 solver threads
Method	2 (barrier)
Crossover	0 (off)
NodeMethod	2
BarHomogeneous	1
BarOrder	0
BarIterLimit	3000
DualReductions	0
ScaleFlag	2
Threads	10

Table A3: **ORACLE configuration** of the final run (run 15).

Setting	Value
Cost slack ε	0.1
Convergence tolerance	0.1 (normalised L_∞)
Maximum iterations	700
Cost coordinate	included (10 explored dimensions)
Step-2 formulation	strong-duality single level (bilinear, non-convex QP)
Step-2 solver options	MIPGap 0.02, MIPGapAbs 0.01, TimeLimit 120 s, Threads 10, NonConvex 2
Final certificate time limit	43 200 s (12 h)
Initial metric cap	1.0 (normalised units)
Cut-validity trigger	10^{-4}

Table A4: **Settings** in which the three reported **ORACLE runs** differ; all other settings are shared and given in Tables A1, A2 and A3. Run 8 predates the **corrected metric readout** (Section 3.2); run 13 was certified post hoc, outside the run (Appendix A1.3).

Setting	Run 8	Run 13	Run 15
Convergence tolerance	0.02	0.25	0.1
Maximum iterations	1200	200	700
Step-2 formulation	KKT-MILP (SOS1)	KKT-MILP (SOS1)	strong duality
Metric readout	incumbent (uncorrected)	proven bound	proven bound
Step-2 time limit per solve	1800 s	600 s	120 s
Step-2 relative MIP gap	0.05	0.05	0.02
Step-2 absolute MIP gap	0.0005	0.005	0.01
In-run final certificate	none	none	12 h

Table A5: Configuration of the **interior sampler** (run 15, rejection route).

Setting	Value
Chains	8, dispersed starting points
Burn-in per chain	2000 steps
Thinning	none (dependence handled via ESS)
Correlation pilot	20 000 steps (sizes the burn-in)
Acceptance pilot	100 000 proposals (sizes the proposal budget)
Target minimum bulk-ESS	10^5
$\hat{R}$ gate	< 1.01 on every axis
Proposal cap	6×10^7
Membership/chord LP tolerances	10^{-9} (feasibility and optimality)
Master seed	42

Table A6: **Exploratory axes** with their exact dataset identifiers, selected capacity type, unit and dataset capacity limit (Section 3.2). The **limit** is the **summed capacity_limit over all nodes for the technology axes** and the **annual import availability for biomass**; “none” marks an axis whose members carry no finite limit. Where the certified near-optimal maximum (Table 1) equals the limit, the **axis is bounded by the dataset rather than by cost**.

Axis	Type	Dataset members	Unit	Limit
Nuclear	tech capacity	nuclear	GW	119.1
Photovoltaics	tech capacity	photovoltaics	GW	10 187.9
Wind offshore	tech capacity	wind_offshore	GW	1149.0
Wind onshore	tech capacity	wind_onshore	GW	4565.0
Electrolysis	tech capacity	electrolysis	GW	none
DAC	tech capacity	DAC	ktCO ₂ eq/h	none
Battery	tech capacity	battery	GWh	none
CCS lump	tech capacity	BF_BOF_CCS, biomass_plant_CCS,	ktCO ₂ eq/h	none
		cement_post_comb,		
		natural_gas_turbine_CCS, NG_DRI_CCS, SMR_CCS		
Biomass import	carrier import	biomass	GWh	777 923.7
Cost	cost coordinate	net_present_cost	MEUR	—

A1.2 Full **technology–carrier matrix** of the Crystal Ball model

Figure A1 lists **every technology (row) and energy carrier (column)** of the modelled system, marking for **each technology the carriers it consumes, produces, or references**, and which carriers may be imported or carry a final demand.

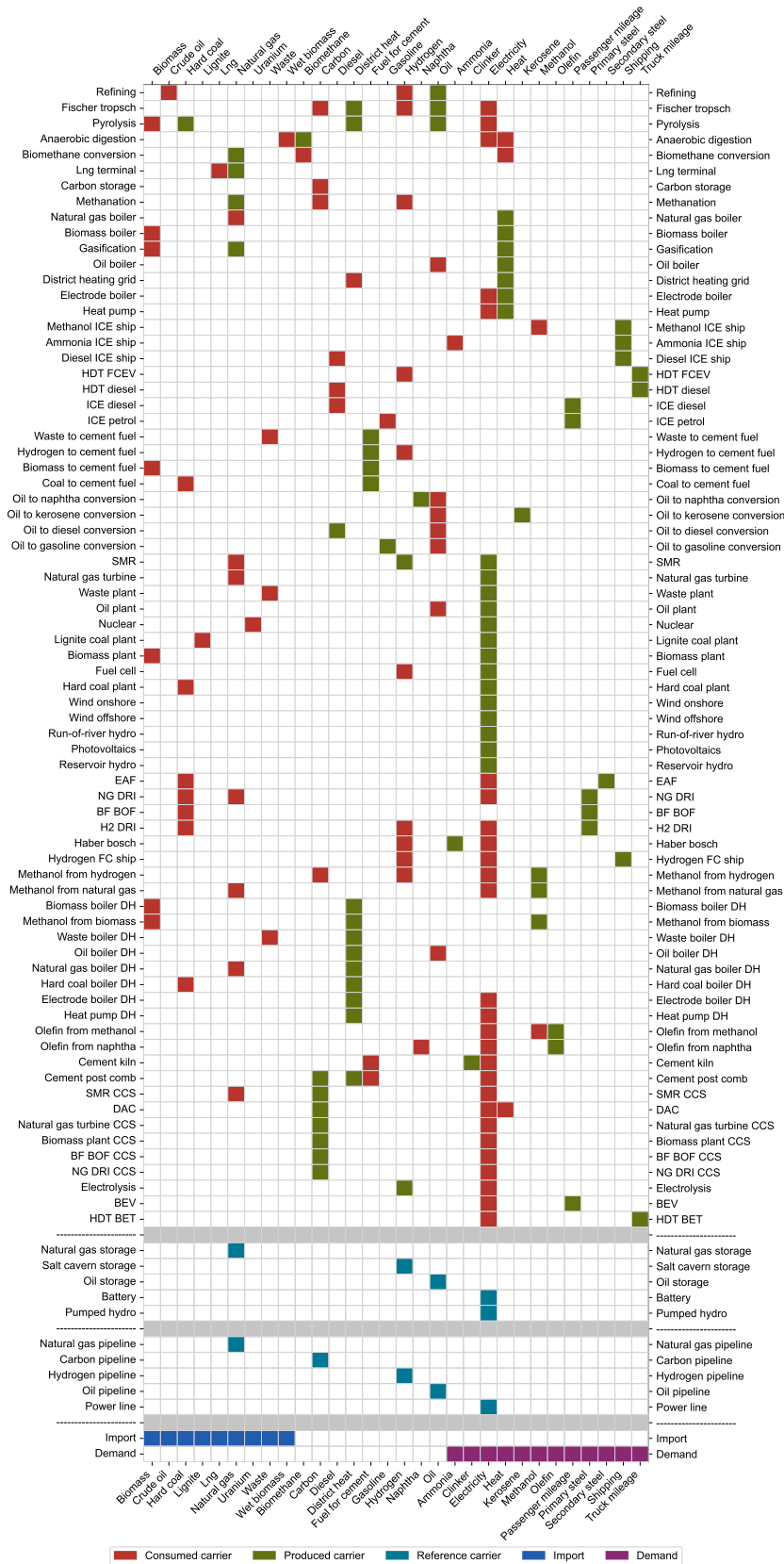


Figure A1: Technology-carrier matrix of the modelled energy system: a red (green) cell marks a carrier consumed (produced) by a technology, a turquoise cell a storage or transport reference carrier, a blue cell an importable carrier, and a purple cell a final demand. Reproduced from the supplementary material of [24]. The figure shows 83 of the 84 technologies and 30 of the 31 carriers in the dataset used here; the modelling helper pair `industrial_gas_consumer` (technology) and `natural_gas_industry` (carrier), which together represent exogenous industrial gas demand, are omitted.

A1.3 Certificate formulations for the convergence metric

Figure A2 compares the **two single-level reformulations** of the **step-2 max-min problem** (Section 3.2) on the fixed run-13 geometry, each given a 6-hour budget. It is the **evidence behind the choice of the strong-duality (bilinear) formulation** over the published KKT-MILP for certifying convergence.

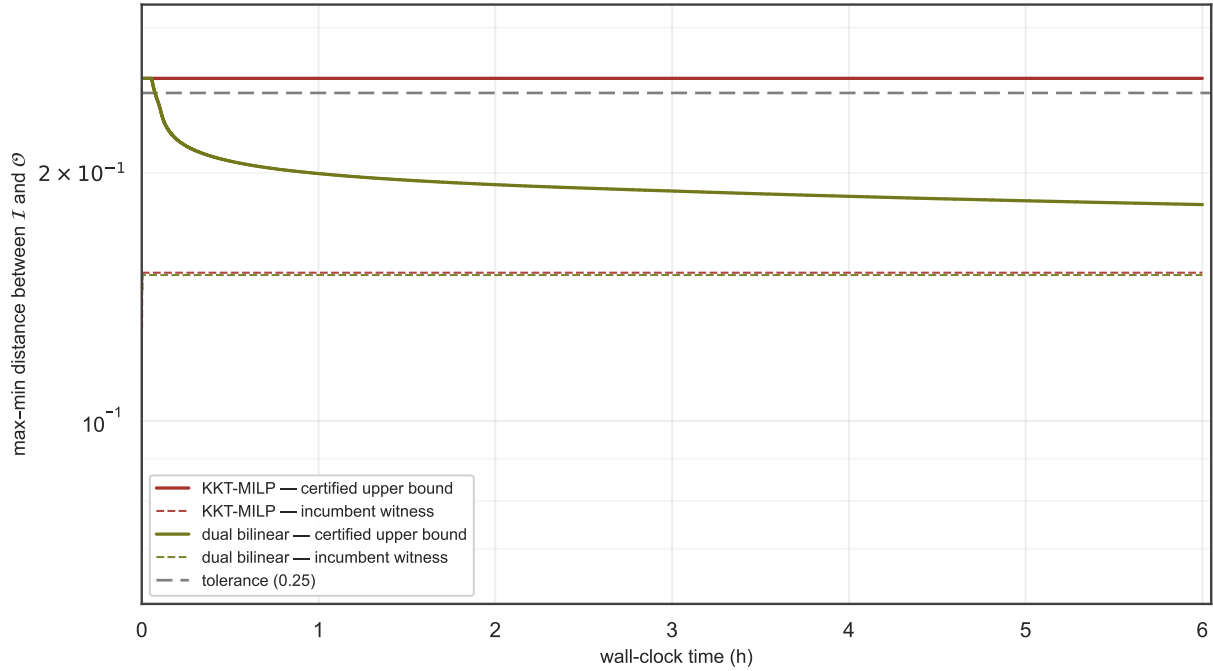


Figure A2: The two certificate formulations of the max-min distance between $\mathcal{I}$ and $\mathcal{O}$ on the fixed run-13 geometry, over a **6-hour budget** each. **Solid**: the **certified upper bound (proven bound)**; **dashed**: the **incumbent (lower bound)**; **grey**: run 13's **convergence tolerance (0.25)**. The KKT-MILP proven bound never improves on 0.260, whereas the bilinear strong-duality formulation closes the bound to 0.183; both **incumbents reach their final value within the first seconds and read as flat**. The certificate brackets exactly the distance the ORACLE metric estimates (same quantity as Figure 5).

A2 Supplementary **interpretation** material

Supporting views of the **run-15 near-optimal sample set and geometry**, referenced from Section 4.3: the **scatter and exact-geometry** companions of the **density matrix** (Figure 6), the **dependence analyses** that justify reading the **pairwise couplings through Pearson correlation alone** (Figure 7), and the **PCA** of the draws.

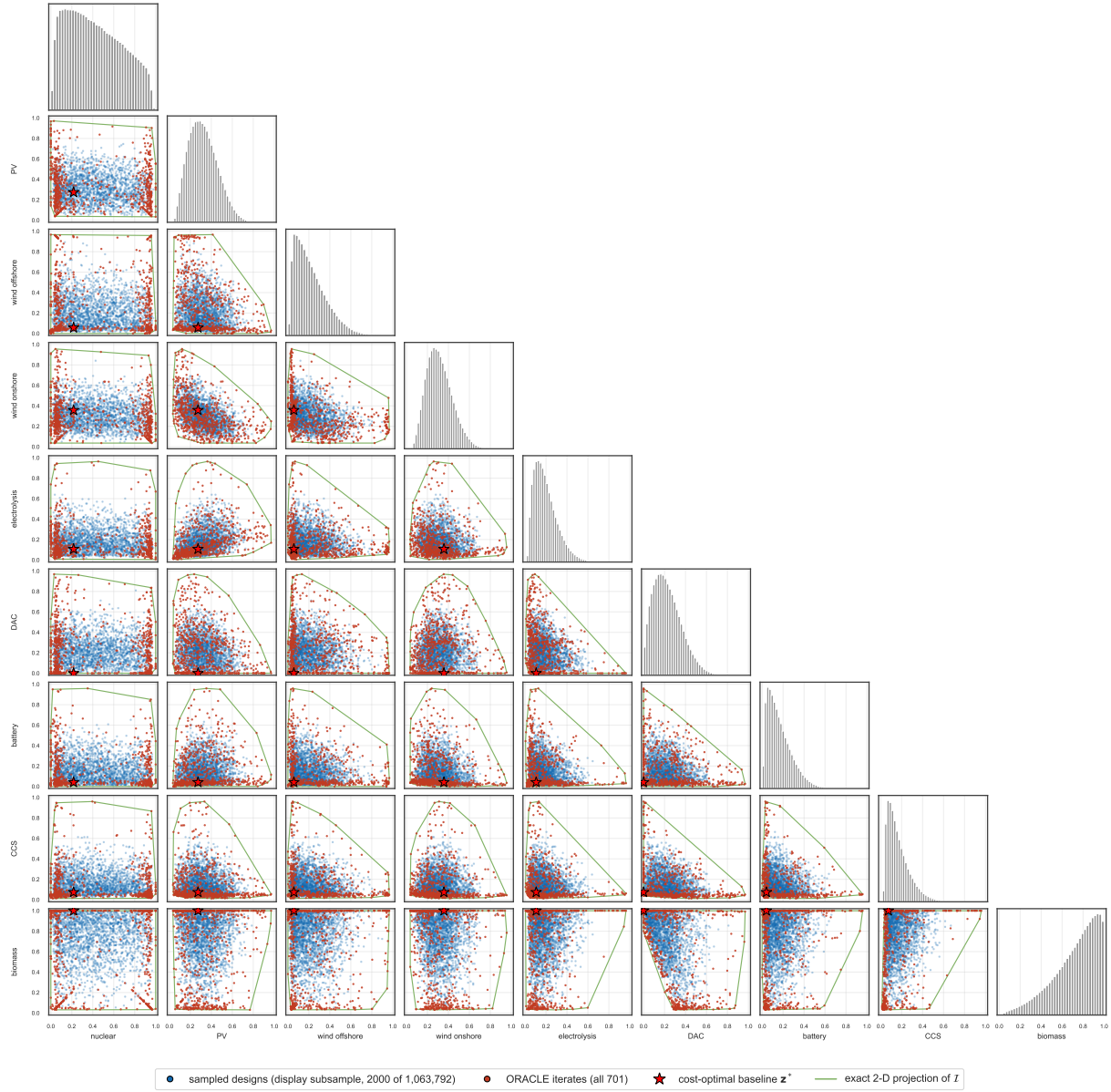


Figure A3: **Scatter** matrix of the **interior samples** over the nine design axes (run 15; axes normalised by their near-optimal maxima). Lower triangle: a seeded 2000-draw display subsample (blue), with all ORACLE iterates (red) and the cost-optimal design $\mathbf{z}^*$ (star) overlaid, and the exact 2D projection of the inner approximation $\mathcal{I}$ outlined in green; diagonal: marginal histograms of the full sample set.

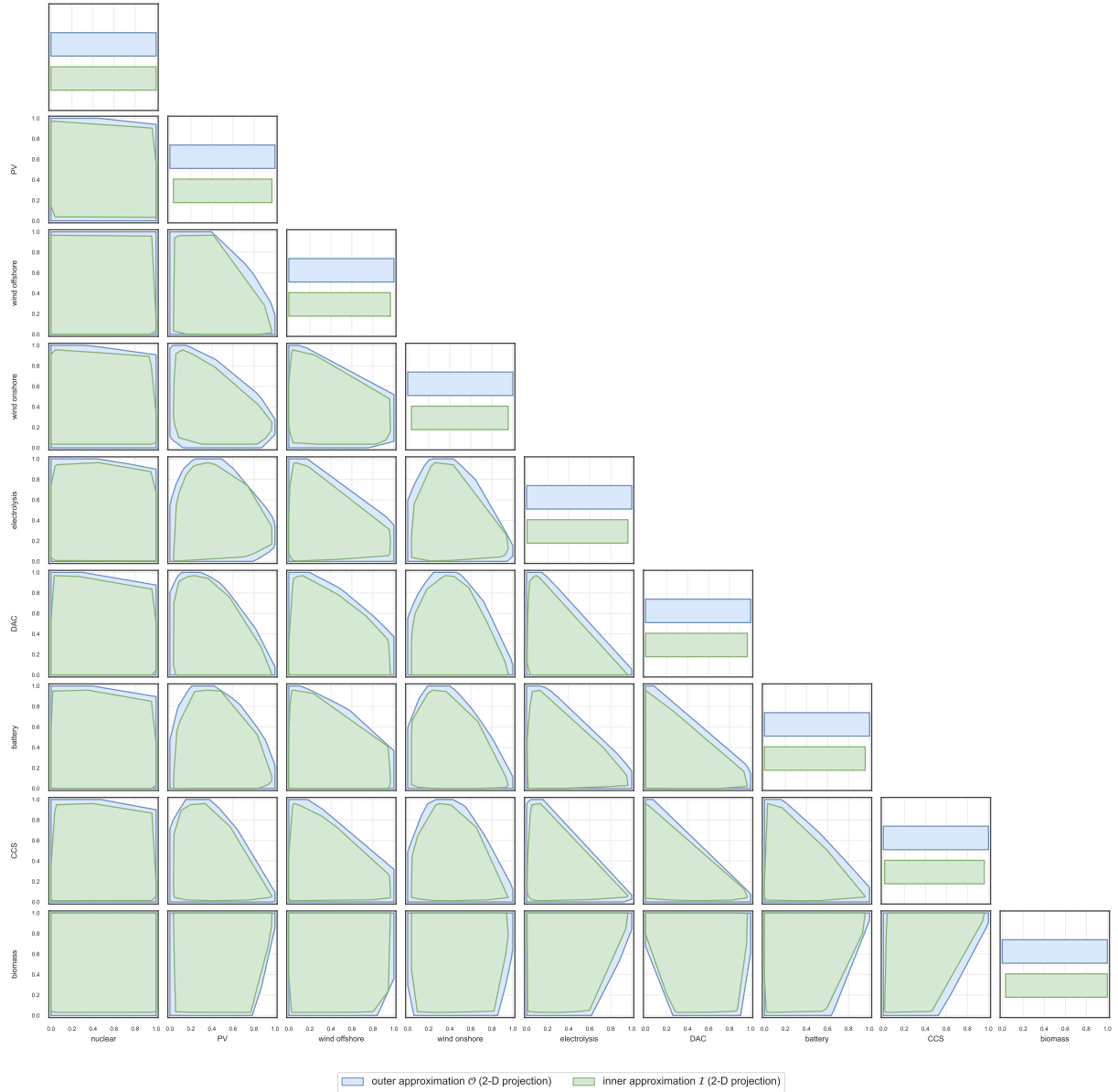


Figure A4: **Projection matrix of the two approximations** (run 15, no samples): for every axis pair, the 2D projection of the outer approximation $\mathcal{O}$ (blue, an **inscribed polygon from 180 support-direction LPs**) containing the 2D projection of the inner approximation $\mathcal{I}$ (green, the **hull of the projected certified points**); the **true** near-optimal region lies **between** them. Diagonal: the **per-axis 1D ranges of both bodies**. For a convex body, the conditional envelope of one axis swept over another is exactly the boundary of that pair's 2D projection, so this figure carries the full envelope information.

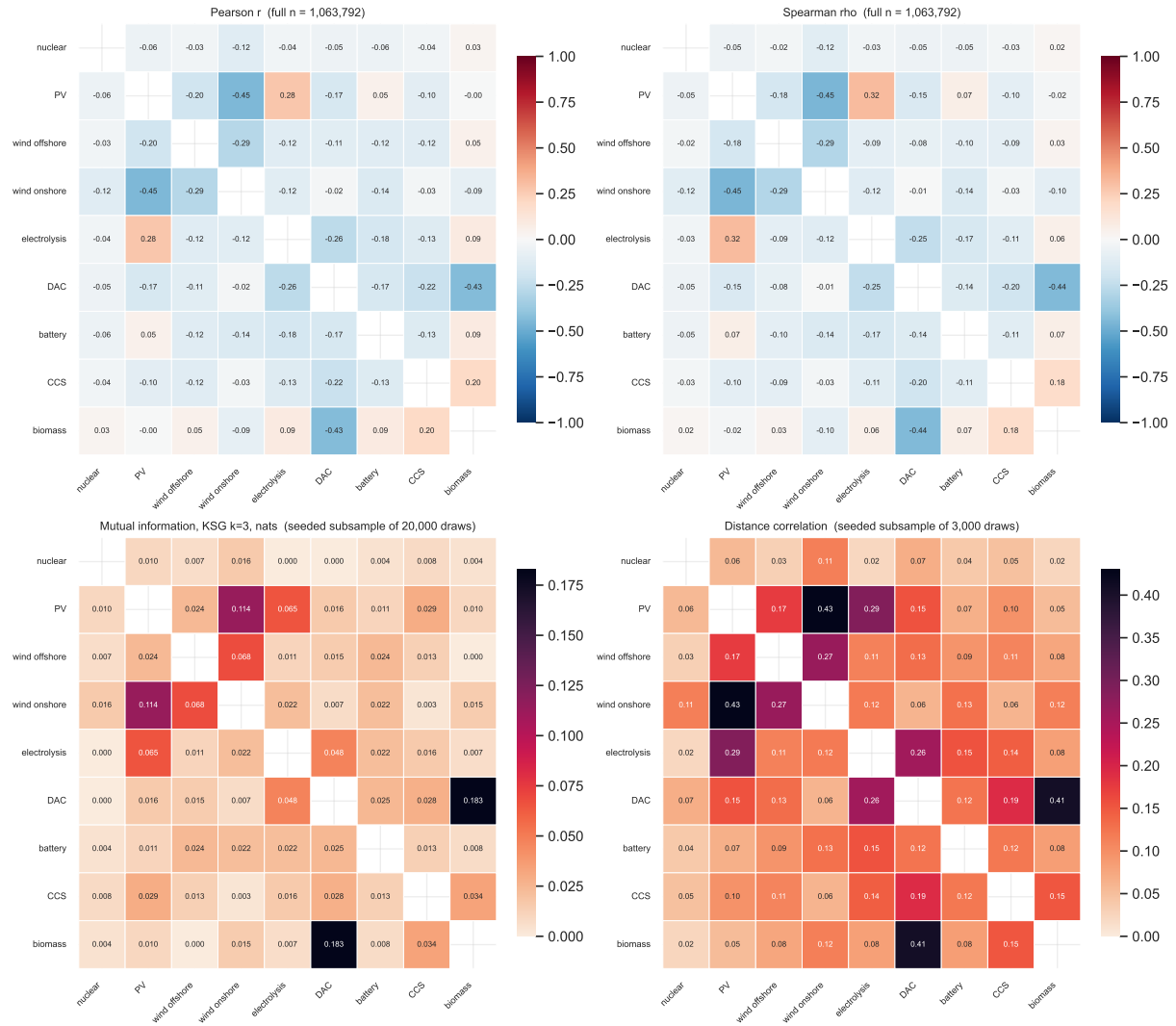
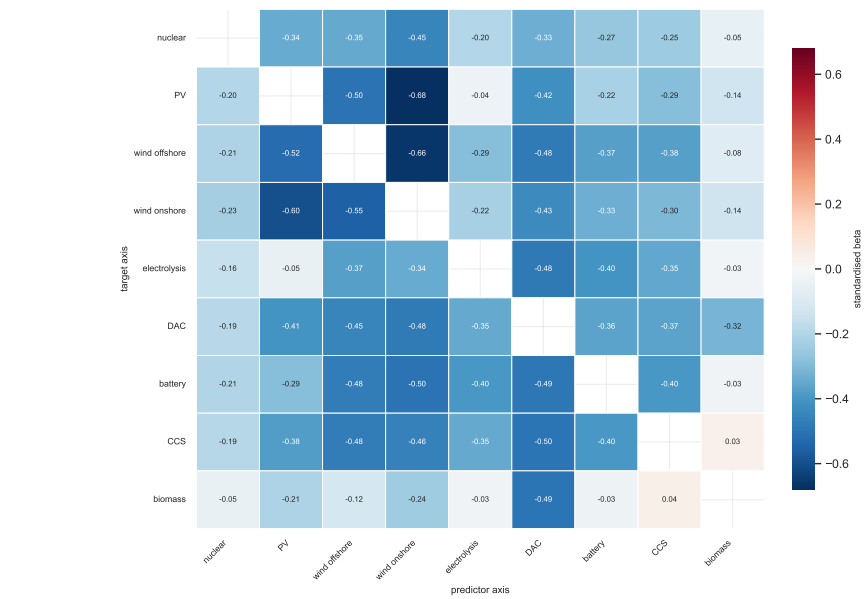


Figure A5: Pairwise dependence across the nine design axes (run 15): **Pearson and Spearman correlation** (full sample set), KSG **mutual information** ($k = 3$, 20 000-draw subsample) and **distance correlation** (3 000-draw subsample). **Spearman $\approx$ Pearson** (no monotone structure beyond the linear one), and neither mutual information nor distance correlation reveals any pair with appreciable dependence that the linear screen misses — together the justification for using Pearson alone in the main text. Diagonals are left blank: for mutual information the **diagonal (self-information) is unbounded for continuous variables**, so all four panels omit it for consistency.



(a) Standardised coefficients

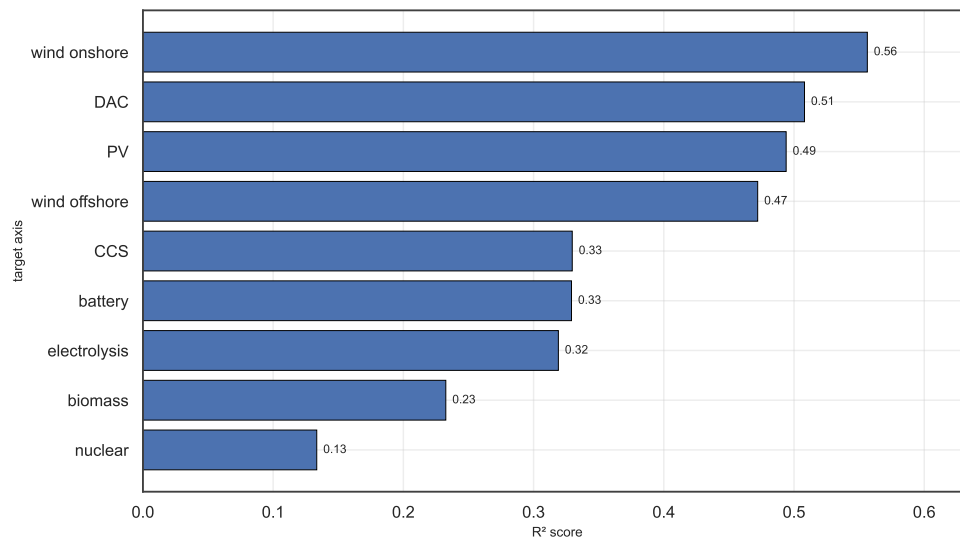
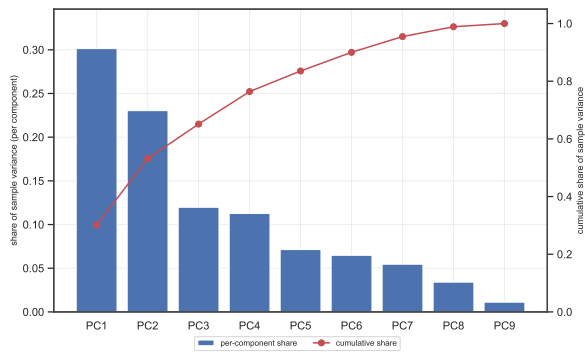
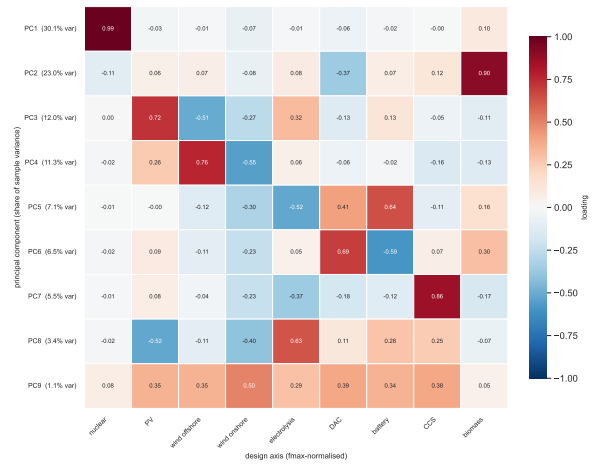
(b) Per-axis R^2

Figure A6: Round-robin OLS on the z-scored axes (run 15): (a) the standardised coefficients of each axis regressed on the other eight, and (b) the per-axis R^2 , i.e. how strongly each axis is linearly pinned by the rest (linear coupling, not remaining feasibility).



(a) Explained-variance spectrum



(b) Component loadings

Figure A7: **PCA** of the normalised draws (run 15; eigen-decomposition of the sample covariance of the raw normalised coordinates, deliberately not re-standardised so the shared $[0, 1]$ scale is retained [44]): (a) **variance per component** with the cumulative share and (b) the component loadings, each row an eigenvector of the sample covariance. **A flat spectrum indicates the space is not dominated by a few trade-off directions.**

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